

AQI Tower — consolidated engineering handoff

D01 / R03M mechanical / E05 control-module fit / 5 October 2026

Decision: ready to hand to the internal engineering team for a build-release review. NOT ready to order all parts, fabricate unchanged drawings or energize. Physical prototypes built: 0.

Use one current design

Four ARCTIC P14 Max ACFAN00287A fans and two proposed IKEA104.633.30 filters in parallel, in the preserved R03M rectangular cabinet. External12V Noctua adapter/controller/hub are the reference power chain. The larger cylindrical outdoor tower is a later project, not this build.

What is now closed digitally

Evidence	What it establishes — and does not establish
Mechanical CAD	493 original assembly objects, STEP and15 drawing files retained. Geometry checks do not approve materials, guards, seals or stability.
E05 control fit	Same300x200x120mm enclosure. Hub rotated/moved, speed controller moved, buttons at X=-15/45/105,Y=55mm. Unchanged assumed rear/cable reservations now clear each other and OEM box geometry. Actual connectors, button stack and supports are unverified.
Software	1,041 host assertions; default Opta target compile/link passed with official core4.6.0. All relay outputs remain disabled. No physical test or protective control implemented.
Pressure and power	At assumed300m3/h total, conditional filter allowance14.316Pa each. Four fans catalogue16.8W; plus Opta max2W gives18.8W before hub/controller/auxiliaries/startup. The24W path limit is not a measured margin.
Mechanical batch	Actual43mm closing gap /36mm sleeve /7mm offset extracted. Robust seal-window solver and four eight-bolt joint-demand groups,10 tests. The broad ASSUMED tolerance case has no valid fixed stop; gasket force/material capacity remain unknown.
Electrical batch	24 power /36 wire-drop /8 input scenarios,8 tests. Fan startup multiplier ceiling1.3095x before auxiliaries; an ASSUMED2A-limited fault cannot establish clearing of a2A fuse. Actual fault/inrush data and protective circuit unselected.
Current parts / tools	28-line current parts register separates exact references/candidates from UNSELECTED items; historicalIH02 retained. One batch-verification command and exact-byte pressure provenance. Visuals/room model remain explanatory, not CFD/CADR proof.

Control decision

START/STOP/RESET are ordinary inputs. STOP is not emergency stop or isolation. RESET does not start fans. Internal speed adjustment is a disconnected-source preset; close the box before operation. No live knob modification proposed. Current OEM Digest19-42 confirms low-power contacts but gives no numerical minimum switching range.

How to read this combined PDF

Pages1–2: current decisions and real release blockers. Pages3–4: E05 layout/input review. Pages5–6: preserved E02 bench-control drawing. Pages7–56: earlier50-page engineering handoff, retained as historical evidence. Where its fit/control or board-build status differs, the E05 check record and current target-build JSON govern; older wiring is not released.

What must happen before construction

These are missing engineering inputs and physical acceptance steps, not items that can be marked complete by generating another document. No build approval has been given.

Five remaining release gates

Gate	Exact deliverable needed / responsible scope
1. Guards / structure	Mechanical reviewer and fabricator: guard reach/strength and seams, material grades, joints, feet, whole-device mass/stability and handling. Revised signed detail drawings and unpowered inspection evidence.
2. Filters / harness	Exact delivered filter tolerance and gasket/compression evidence; fan/extension lead, connector and clamp geometry. Resolve service retention, bypass seal and protected entries. OEM enquiries already sent; replies pending.
3. Rated electrical design	Electrical reviewer: hazard-to-protective-function allocation including restart/recovery; complete selected protection/isolation/switching, startup/fault coordination, actual gauges/terminals/glands, metal-panel treatment, mounts and thermal assessment. Opta software cannot replace this.
4. Actual assembly / commissioning	After revision-specific release and authorized procurement: supplier delivery, fabrication, dry-fit inspection, safe qualified assembly, isolation/stop/restart and abnormal/thermal tests. Signed commissioning precedes performance tests.
5. Performance / acceptance	Test lead: agreed indoor room/targets, calibrated or paid/rented instruments. Record airflow, pressure, bypass, noise, power, temperatures and controlled PM decay/background. No proven CADR or outdoor bubble radius yet.

No-lab construction route

Use a local cabinet/metal fabricator and qualified electrical assembly/test service in Pune, Hyderabad or Mumbai when chosen. College/NGO funding supports those services and approved parts; no college workshop is assumed. Prepare quotations from current drawings clearly marked REVIEW ONLY; production follows signed release, not this PDF.

Critical path — no invented dates

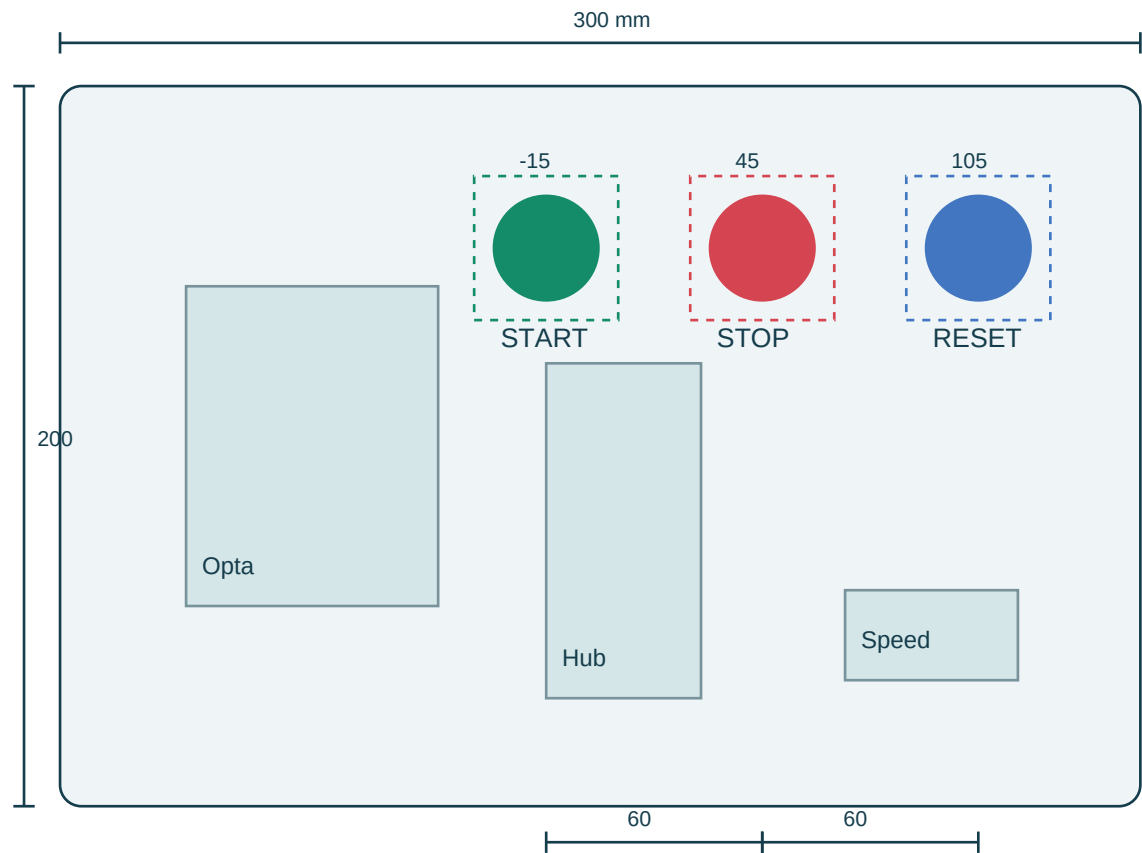
Resolve exact component inputs and protective functions → signed mechanical/electrical revision → funding and purchase authorization → delivery/fabrication → unpowered fit and assembly → qualified commissioning → measured performance. Supplier replies and workshop/test capacity control elapsed time; digital compilation alone does not shorten them to a promised date.

Funding and technical sheets

STAKEHOLDER_FUNDING.xlsx is for actual quotations and milestone funding. TECHNICAL_REVIEW.xlsx and CURRENT_RELEASE_GATES.csv preserve decisions and closure evidence. Stock, prices and dates are UNKNOWN until confirmed. Only released line items can enter a purchase/build authorization.

Unchanged boundary: indoor experimental particle demonstrator. No gas-removal, whole-device HEPA, medical, outdoor heat/rain/tamper or public-clean-air coverage claim. No purchases, supplier messages, hardware flashing or energization performed in this continuation.

A closed lid with three clearly labelled controls



Datum: box centre X/Y; proposed row Y=55; not a drilling template
Dashed squares: ASSUMED 40 x 40 rear envelopes

Verified OEM mounting instruction: BRU46063 page2 specifies 22.3mm hole with +0.4/-0 tolerance; 1-6mm panel range for these ordinary heads. Same-row minimum 30mm; proposed 60mm pitch exceeds this. Labels and installed accessories must still fit.

Detected in the private box STEP: lid thickness 4.0mm at all three proposed centres. It lies within the head mounting range. This does not establish delivered thickness, retention or ingress protection after drilling.

Assumed, not measured: rear stack 40x40x55mm below inner lid plus 20mm wire-tail depth. Reservations remain the same size. OEM box collision checks exclude only the mounting panel for supported bodies. Actual button assembly and wire bends still need verification.

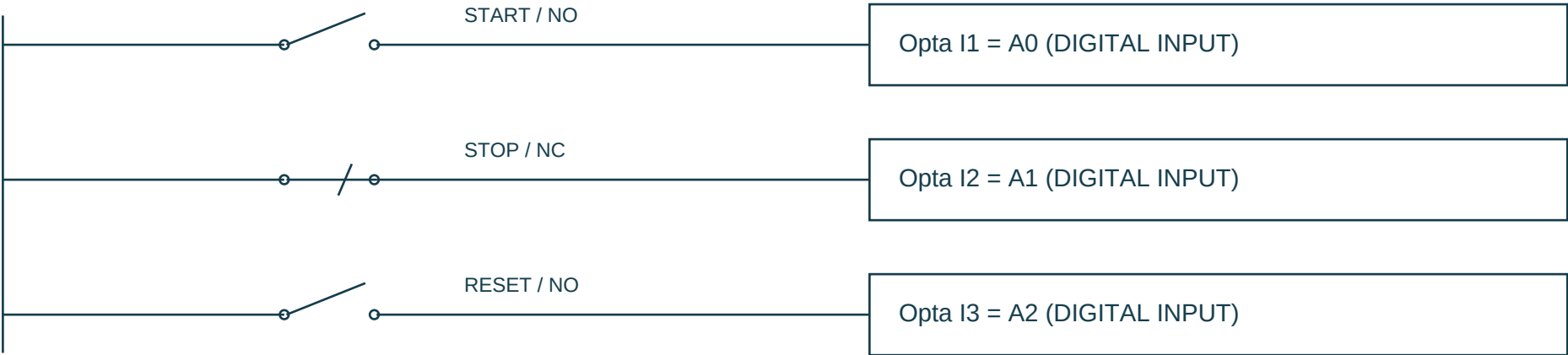
Digital conflict closed: hub rotated 90 degrees and moved; controller moved below button row. All unchanged wire-tail and service reservations now clear each other and the OEM box. Current lower-left coordinates: hub -15/-70; controller 68/-65; buttons Y55. Supports are not designed; see CURRENT_CONTROL_PLACEMENT.csv.

Speed setting: retain the unmodified internal NA-FC1 for now. Disconnect the external source before access; set speed, then close the lid before operation. Live adjustable external knob mounting is NOT designed. Ordinary STOP is not isolation.

Control	Candidate head / collar / contact	Centre X/Y mm	Function
START	ZB5AA3 / ZB5AZ009 / ZBE1016	[-15, 55]	NO spring-return
STOP	ZB5AA4 / ZB5AZ009 / ZBE1026	[45, 55]	NC spring-return
RESET	ZB5AA6 / ZB5AZ009 / ZBE1016	[105, 55]	NO spring-return

Ordinary input circuit - not a safety circuit

PROPOSED PROTECTED +12 V CONTROL BRANCH - PROTECTION / TERMINALS UNSELECTED



Contact screw IDs intentionally omitted - not verified

Input state	Ordinary software interpretation
START NO pressed / I1 HIGH	Request RUN only when existing logic permits. Held START cannot auto-restart.
STOP NC healthy / I2 HIGH	Pressing STOP opens contact; LOW requests stop. This is NOT protective isolation.
RESET NO pressed / I3 HIGH	Clear ordinary latched fault only when logic allows; does NOT itself start fans.
Common / additional inputs	Opta negative shares reviewed source0V. I4 bench permit / I5 source present remain E02 references; not protective inputs.

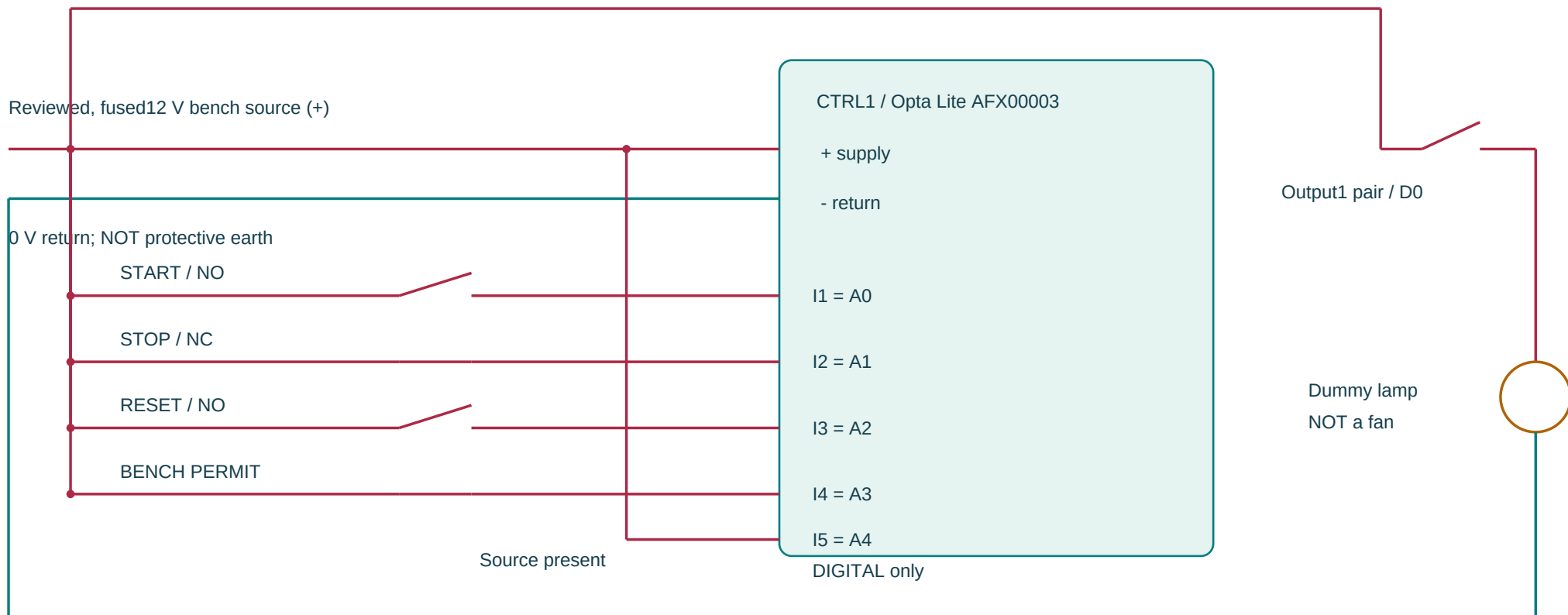
OEM electrical facts: Opta digital input0-24V, HIGH minimum6.6V, LOW maximum4.46V, impedance8.9kOhm,1.12mA at10V.12V/8.9kOhm =1.348mA is a nominal resistance screen only, not guaranteed current. Gold-flashed ZBE1016(NO) / ZBE1026(NC) are manufacturer-designated low-power contacts. Their numerical minimum switching range at our conditions remains UNKNOWN.

Bench-only status: default sketch leaves every relay OFF; this circuit cannot yet operate the fans. No protective PR1 selected, no fan-power contact assignment, no fuses or wire sizes released. Verify contact minimum rating, terminal IDs, retention, source/branch protection, actual cables, service loop, heat and isolation before any dummy-load commissioning.

Primary sources checked5 Oct2026: Schneider BRU46063 page2; Harmony catalogue; current Digest0100CT2401-SEC-19 page19-42; Arduino Opta input specification. Full links in module JSON / README. Current Digest confirms low-power parts but does not give numerical minimum switching limits. OEM files stay private. Physical prototypes0; R03M unchanged.

A real controller and implemented local sequence

Terminal reference for a protected dummy-load bench only. PR1 and the tower protective system remain OPEN.



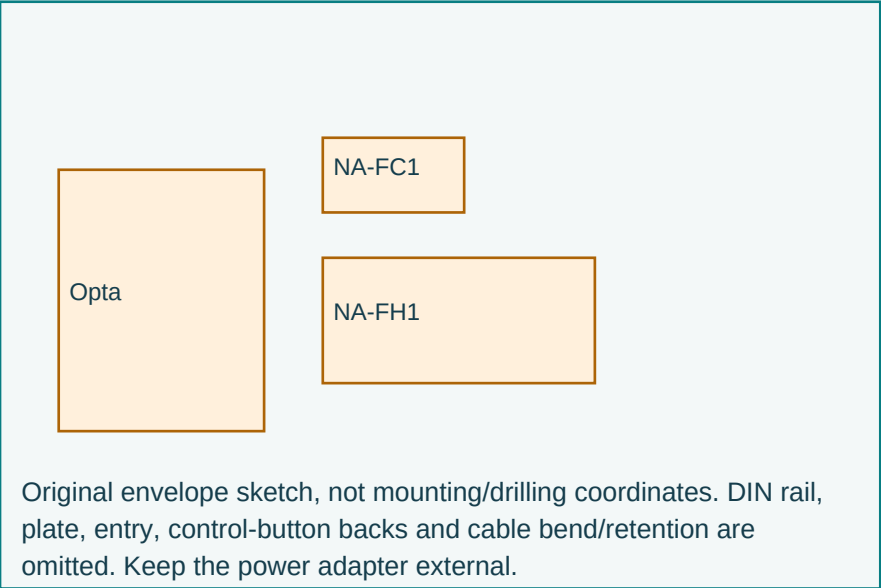
Connection hold: junction dots connect; crossings without dots do not. Contacts and power protection are not approved parts. Output1 is its printed contact pair, not invented COM/NO identifiers. Preserve polarity; compatible protected harness required. No connection to the tower fan chain is released.

Firmware: all relay commands remain LOW by default. RUN REQUESTED/TRIP/source LEDs show software status only. Inputs I1–I5 are digital. 12 V is outside the 0–10 V analog range; do not change I5 to analog mode. I4 is a bench test signal, not independent protective permission.

Enclosure fit and protection coordination

A concrete external-box candidate; not a fabrication layout, released fuse choice or completed safety circuit.

Hammond1554XA2GY /300 x200 x120 mm



Fit finding: Opta front70 x88.8 mm; drawing-derived depth reservation61.1 mm. A40 mm deep CAD box is not a normal DIN fit. Do not reinterpret the stock box IP rating as modified-device weather protection. The new external box is a recommendation; native tower CAD is unchanged.

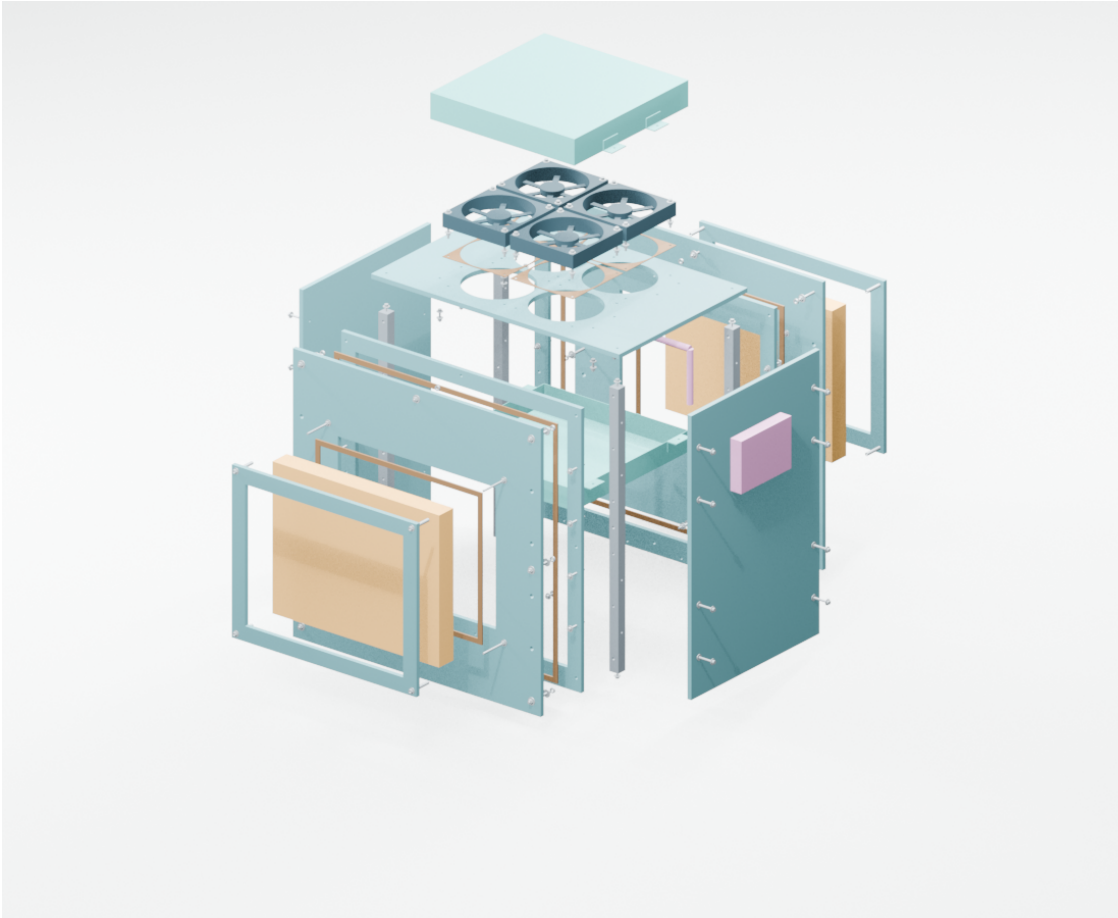
Power: Opta maximum2 W at12 V adds0.167 A. Fans plus Opta alone total1.567 A /18.8 W. Supply margin0.433 A remains before hub/controller, input circuits and startup. No fan power-budget approval follows.

Real fuse screen	Verified OEM data	Decision / consequence
Littelfuse0297002.WXNV	2 A /32 V DC;1000 A interrupt rating at32 V	SCREENED, not selected. Holder/actual circuit coordination unresolved
Typical thermal allowance	1.9 A at20 C;1.7 A at40 C	Fans+Opta leave only0.133 A to40 C allowance before other loads
Opening at110% /2.2 A	Minimum360000 s; no maximum given	Not a2 A current limiter or proof of prompt clearing on a limited supply
Opening at200% /4 A	0.15–5 s specified window	Actual PSU/hub fault current/time and weakest cable/contact required

Limits: 1,041 host assertions pass. Default sketch compiles for Opta (official core4.6.0 / CLI1.5.1). All physical tests are NOT DONE. Software, ordinary relays, digital source presence and LEDs do not prove safe access, fault coverage or standstill. Existing restart rules remain unchanged and unresolved for internal fan/hub resets.

AQI Tower Engineering Handoff

Current D01 R03M integration with IH02 evidence



NOT FOR FABRICATION OR ENERGIZATION. Authoritative manufacturing geometry remains the FreeCAD R03M study; Blender and browser meshes are display derivatives.

The following review pages lead into the unchanged 41-page IH02 engineering reference. Return named decisions with evidence in TECHNICAL_REVIEW.xlsx. Do not mix legacy large-tower mains/HEPA assumptions with this 12 V indoor demonstrator.

Current pressure and geometry

Quantity	Current screen or evidence
Total flow	300 m3/h assumed, not measured
Parallel branches	4 fans at 75 m3/h each; 2 filters at 150 m3/h each
OEM fan head	27.63 Pa at stated curve condition
Nonfilter loss	13.31 Pa under K1 reducer scenario
Remaining filter head	14.32 Pa per parallel filter branch
CAD inventory	493 source objects; tessellated display derived from saved R03M
Display bounding envelope	595 x 504 x 631 mm including reservations and projecting hardware; not a fabrication specification

The two parallel filters do not add their pressure drops in series. Clean/loaded filter curves, installed/static fan curve compatibility, loss coefficients and branch equality remain unresolved. This screen is not an operating-point prediction.

Colour and animation interpretation

Teal identifies cabinet/guards, amber filters, dark blue fan envelopes, purple reserved control space. Motion shows part separation or hand-authored air paths, not velocity, turbulence, deposition or removal. Guard faces remain solid tessellations, not the proposed perforation geometry.

Review responsibilities and construction path

Discipline	Closure required
Mechanical	Guard strength/reach, seams and attachments; cabinet joints and stability; real filter tolerance, gasket compression, safe service access and whole mass/CG.
Electrical	Hazard allocation, independent protective functions, actual DC switching/isolation/protection, wire/connector ratings, cable restraints and enclosure. No guessed terminal assignments.
OEM-dependent	Exact fan/extension connector and clamping dimensions; delivered filter/frame properties. Sent enquiries are pending.
Commissioning and test	Confirm premises and targets; review assembly, isolation/restart and abnormal behaviour before collecting measured performance.

Critical path

Exact component evidence and accepted targets -> mechanical/electrical review -> revision-specific fabrication release -> supplier delivery and fabrication -> assembly and qualified commissioning -> airflow/pressure and controlled particle tests. These dependencies are not completion dates.

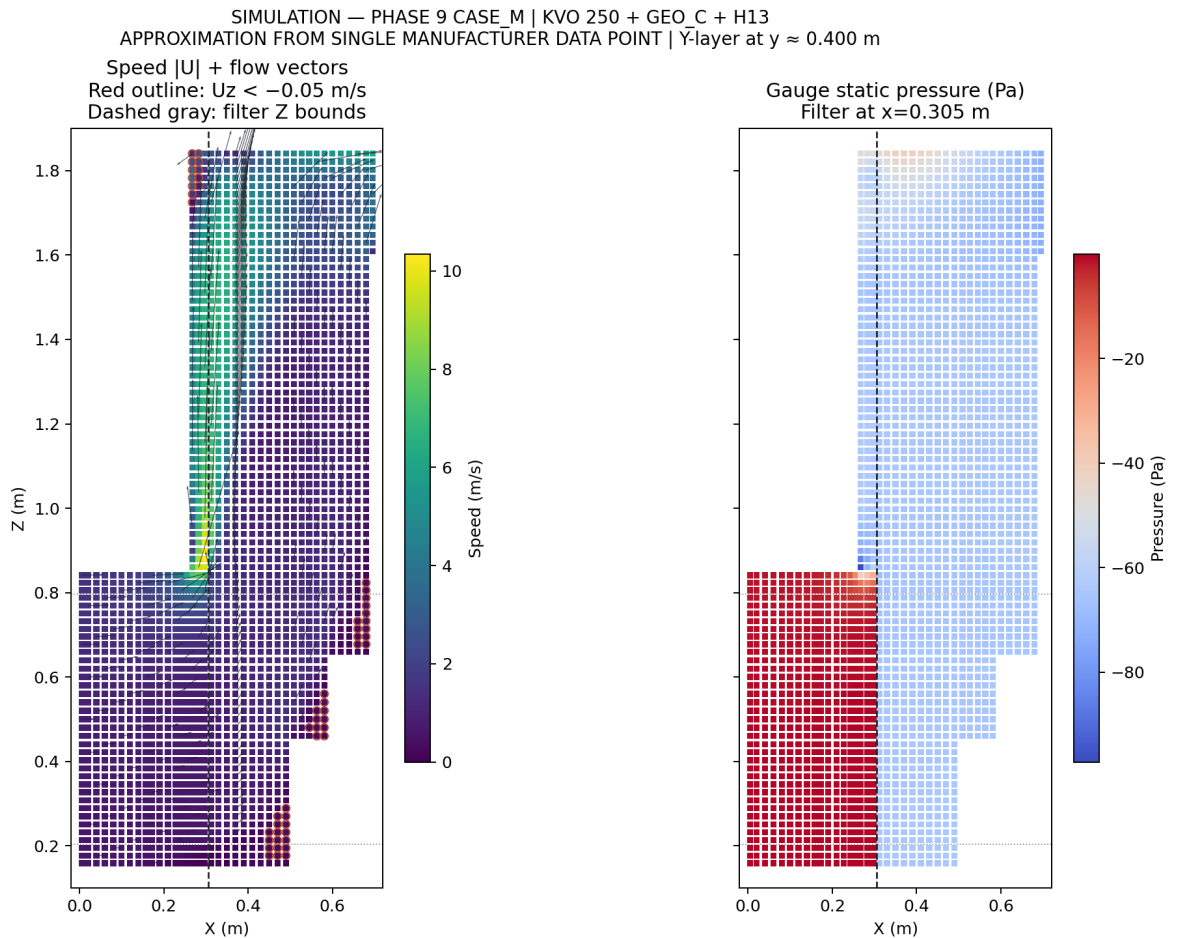
Outdoor boundary

Current fan temperature limits, enclosure and materials do not establish suitability for Indian outdoor heat or public misuse. Weather protection, tamper-resistant service access, anchoring and wind-dependent performance require a separate qualified design and approved site.

Files to use

Native CAD, STEP, 15 DXFs, current tools, measurement templates and source registers are in engineering/. TECHNICAL_REVIEW.xlsx combines parts, release, pressure and hardware sheets. The Blender file is for presentation only.

Historical CFD evidence



Existing Phase 9 CASE_M plot: KVO 250 + GEO_C + H13 approximation from a single manufacturer data point. Colours show calculated speed and gauge pressure for that historical case, not R03M performance.

GEO_C is an earlier air-path geometry trial. It is not the current manufacturing assembly. Historical CFD is partial/nonconverged; the later numerical bypass seal is not a proven physical seal. No new R03M CFD has been run for this delivery.

For the current prototype, use the pressure screen and obtain measured filter/installed-flow evidence. The following IH02 appendices are preserved unchanged.

Room duty and validation tools

Digital supplement. T01 remains OPEN: actual premises, accepted targets and instruments are not selected. The viewer calculates scenarios, not D01 CADR. First scope remains airborne particles; no gas, medical or outdoor coverage claims.

Mass balance and performance requirement

Volume $V = \text{area} \times \text{ceiling height}$. Ideal particle concentration ratio = $\exp(-\text{CADR} \times t / (60 \times V))$; time in minutes = $-60 \times V \times \ln(1-\text{reduction}) / \text{CADR}$. CADR is size-dependent effective particle-cleaned flow, not fan free-air flow or the 300 m³/h pressure-screen assumption. Source/influx, settling, ventilation and nonuniform mixing are excluded.

Illustrative room at 2.8 m height	90% in 30 min requires CADR	Ideal 90% time at assumed 150 m ³ /h
10 m ² / 28 m ³	128.9 m ³ /h	25.8 min
25 m ² / 70 m ³	322.4 m ³ /h	64.5 min
50 m ² / 140 m ³	644.7 m ³ /h	128.9 min

For a filtration-only mass balance, effective clean-air delivery cannot exceed actual installed airflow at 100% capture. The 25 m² example needs more than the assumed 300 m³/h screening flow. This is a conditional feasibility limit, not proof of delivered airflow. Accept room/target only after measured duty is known.

Tools carried forward

engineering/scripts/analysis/indoor_decay.py and its 8 synthetic tests are now included. This existing single-pair off/on fit estimates conditional additional particle-cleaned flow only if its assumptions are explicitly declared. It is not certified CADR; repeatability and uncertainty remain unevaluated. Preserve calibration, source/background, ventilation state, timing and measured room volume. Use the existing recorder schema, not the older PARTICLE_TIMESERIES_BLANK.csv interchange format.

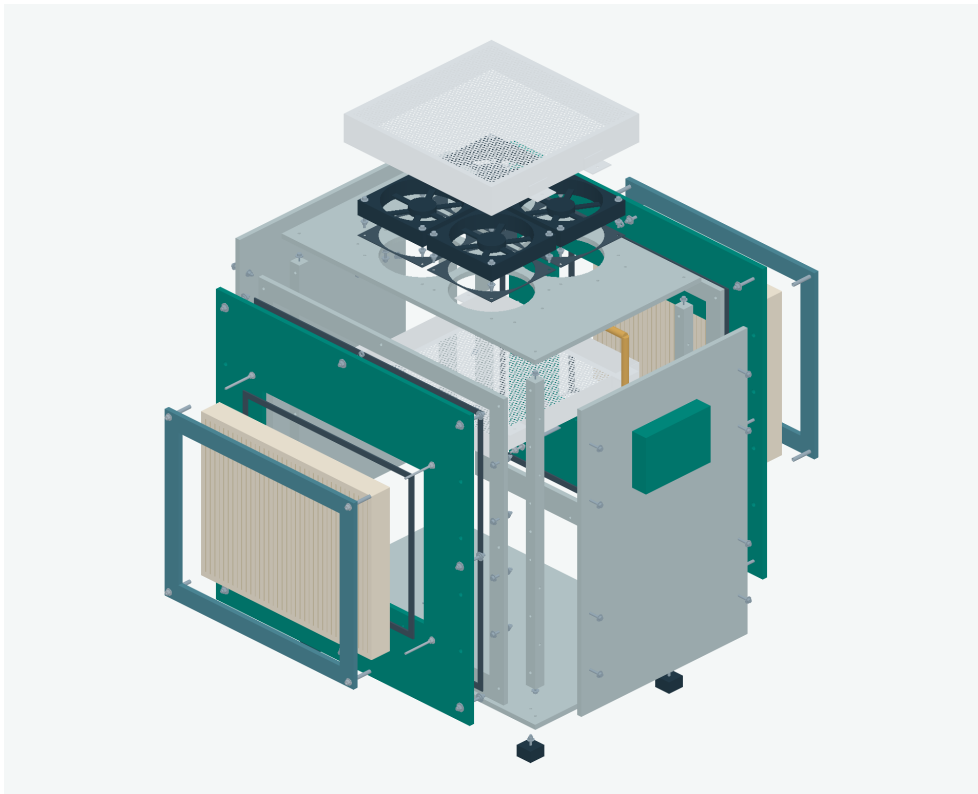
A new room-scenario workbook tab and ROOM_MODEL_BASIS.md explain inputs, limitations and sources. The interactive person and room are visual references, not a modeled occupied ventilation field. No guard, electrical or structural approval has been granted by this supplement.

Primary references checked 5 October 2026: EPA Guide to Air Cleaners in the Home; CDC Appendix B Air. See ROOM_MODEL_BASIS.md for exact links and applicability limits.

D01 Internal Engineering Handoff

Review revision IH02 | Mechanical R03M | Electrical E00 with supplements | 5 October 2026

Decision requested: review and resolve the open engineering items in REVIEW_AND_RELEASE_REGISTER.csv. This is a coordinated design-review issue, not a fabrication or energization issue. Manufacturer replies are not the only remaining dependency.



Exploded view of the actual current assembly. Omitted threads, weld beads and perforations limit what CAD checks establish. Nominal component fits do not establish strength, safety or sealing.

Evidence class	Meaning
DETECTED	Recorded files or geometry; not physical measurement
OEM REFERENCE	Published component data; not approval of our integration
ASSUMPTION	Provisional input requiring validation
UNKNOWN / OPEN	Do not order, cut or energize the affected interface

Authoritative files and corrections

Use the native R03M assembly as the base. Harness review CAD is a derivative with assumed cable reservations, not a competing machine. The package preserves source-relative paths under engineering/ so scripts can find their inputs.

Use	Controlling source
Mechanical geometry	mechanical_package/D01_R03M_ASSEMBLY.FCStd and STEP;15 DXF files include10 panels plus5 guard blank types
Electrical	electrical_package PDF, BOM and interfaces plus protective selection supplement; no completed rated circuit
Current pressure	CURRENT_PRESSURE_BUDGET.json generated in this handoff from R03M guard area and archived OEM numeric fan points
Current airflow analyzer	engineering/scripts/analysis/d01_validation.py; mandatory --plane-area-m2;23 checks
Latest open decisions	REVIEW_AND_RELEASE_REGISTER.csv and guard/harness/protective supplements

Superseded assumptions: do not use old Filtrete/MERV13 arrangements, large-tower230V fan wiring, earlier long studs or riveted guard details. D01 currently uses two experimental STARKVIND envelopes and four12V P14 Max fans. No certified whole-device filtration class is assigned.

Pressure correction:14.32Pa per filter at an assumed300m3/h total and reducer K1, not the older14.61Pa. This is remaining model headroom, not a measured filter loss or acceptance margin. Old R01 operating-flow predictions do not transfer to STARKVIND.

Appendices: earlier mechanical18-page, electrical7-page and validation5-page PDFs are included as labeled reference snapshots after this review section. The current notes and scripts supersede their outdated analyzer command/test count and any older pressure interpretation. No design release is implied.

Mechanical review disposition

DETECTED:493 objects in the base CAD;119805 checked non-reservation pairs with no positive-volume clash in the saved generator audit. The panel audit reports119804 because it also excludes the guard pair. These are stored geometry results; native CAD hash is verified in this handoff.

Interface	Review action
Filters and seals	370x290x40 filter envelopes; frame bearing and thickness tolerances unverified. Assumed39/40/41mm thickness changes nominal4mm gasket compression from0/25/50%. Do not cut spacers from that assumption.
Cabinet and supports	Panel/cleat grades and joint capacity not selected. Partial assumed mass12.19kg excludes real fans, filters, controls, feet and seals. Do not call it total mass or approve tipping from it.
Guarding	Independent welded guards remain fitted during filter removal.4mm holes/6mm pitch are proposed geometry, not access approval.22/50mm guard-to-body distances are not blade clearance.
Fasteners	Use the R03M schedule and modeled inventory together. One schedule row groups8 adapter studs plus4 ledge screws; do not read its qty8 as the whole hardware count. Grade, preload and wood bearing remain open.
Service	Two200mm filter withdrawal sweeps passed nominal geometry checks. Hands, tools, retainer handling, sag and manufacturing tolerances still require review.

Guard supplement:36 ideal solid-strip cases and28 arithmetic tests explain thickness/span trade-offs only. They do not model the actual2822 holes, plate action, yielding or weld flexibility. A thicker guard or stiffener is not selected. Review loading criteria and unpowered sample assessment.

Electrical review disposition

Reference chain only: OEM NV-PS1 external adapter -> supplied NA-AC10 -> NA-FC1 -> NA-FH1 four-pin input -> four P14 Max fans. SATA stays unused. Instrument logger power remains separate. This is not a terminal-level wiring instruction.

Issue	Required resolution
Power budget	Four fans at0.35A each give1.4A/16.8W nominal.2A/24W path leaves0.6A before controls and startup. A1.5x fan startup scenario alone is2.1A; this is a sensitivity, not a measured surge.
Protective behaviour	PWM zero and hub fuse are not isolation. Hub self-reset and port1-only upstream RPM cannot establish independent four-fan safety monitoring.
Control candidates	Pilz X5/X9P/s4 screens do not select a protective assembly. X9P leaves0.2W before auxiliaries; s4 needs24V; X5 monitored sequence not established.
Enclosure and harness	140x100x40mm is reserved space, not a selected box. Include walls, terminals, bend radii, cooling, glands and access after circuit selection.
Review output	Issue rated schematic, interface/terminal and cable schedule, protection coordination, enclosure layout and commissioning procedure. Retain or explicitly revise restart requirements through hazard review.

Keep OEM mains adapter intact. No bare-mains design is released. Internal electrical engineer determines applicable Indian requirements, supply/site suitability and required inspection. An external adapter rating does not certify this assembly. No arbitrary fuse size, wire gauge or terminal mapping has been filled in.

The14 named control cases and1024 one-step states are software logic checks only. They do not test hardware timing, welded contacts, sensing coverage, DC interruption, thermal behaviour or safety integrity.

Pressure and measurement basis

Four fans operate in parallel: flow adds, pressure does not. Two filter branches operate in parallel: each ideally sees half total flow; their pressure drops are not added. Unequal branch flows remain possible in real construction.

Total m3/h	Each filter m3/h	Fan Pa	Other losses Pa	Filter allowance Pa
200	100.0	32.60	5.92	26.68
250	125.0	30.09	9.24	20.85
300	150.0	27.63	13.31	14.32
350	175.0	25.27	18.12	7.15
400	200.0	22.83	23.66	-0.83

ASSUMPTIONS:2800rpm OEM curve; density1.2kg/m3; equal fan/filter sharing; guard area0.0354623m2 each; guard K1.5 each, plenum K1.5, installation K1 and reducer K1. Full JSON also gives reducer K0.5/2 sensitivity. Coefficients, static/installed compatibility and any double counting remain unverified. Negative headroom means that scenario has no allowance for a positive filter loss.

No actual STARKVIND clean/loaded curve has been established. Do not promise300m3/h, CADR, HEPA-level removal or room coverage. Fit and pressure evidence are separate gates.

Current airflow tool requires accepted physical rows, unique cells, one run/instrument/method/calibration, ISO timestamps with timezone and independent full-plane area. Equal cell-area sum does not prove correct spatial coverage. Reported error is velocity-only; it excludes area, swirl, fixture effects and time variability.

Run from engineering/: **python scripts/analysis/d01_validation.py**. With real reviewed data: add **--airflow-csv FILE --plane-area-m2 AREA --output NEW.json**. AREA is the measurement plane, not automatically guard open area. Never alter raw observations to obtain a passing result.

Review ownership and release gates

IDs	Owner	Closure package
M01-M03	Mechanical engineer	Guard/structure/seal calculations, selected materials/joints, corrected drawings and inspection plan
M04	Mechanical + electrical	OEM reply interpretation and protected harness integration
E01-E03	Electrical + mechanical	Hazard allocation, rated circuit/enclosure and commissioning evidence
P01 / T01-T02	Test lead + project lead	Filter/duty evidence, agreed room/targets, calibrated methods and physical results
C01	Project lead + engineers	Real quotes and released revision-specific BOM/build authorization
O01	Future outdoor team	Separate public/weather/wind/stability design and field evidence

Complete the CSV register with reviewer name, disposition, date and linked evidence. A row is not closed by ticking a box: the accepted calculation, drawing or physical result must exist. No reviewers or approvals have been invented.

What can happen now: internal drawing/calculation review, applicability/hazard decisions, manufacturability discussions and scoped quotations if separately authorized. These do not require the manufacturer cable reply.

Waiting: cable/connector evidence from ARCTIC and Noctua; Noctua ticket220175. Filter/frame/curve data, site selection and actual physical evidence are separate unresolved inputs, not covered by those enquiries.

Release boundary: only a signed revised package with closed safety-critical holds can authorize affected fabrication/assembly. Digital work, supplier delivery, workshop fabrication, inspection and physical testing are distinct critical-path activities. No completion date is promised.

Assembly and verification route

Sequence below applies only after component-specific release. Use the mechanical appendix for drawing details; unresolved quantities/ratings must not be guessed on the workshop floor.

Stage	Action and hold point
1 Receive and inspect	Record exact fan/filter/adaptor revisions, dimensions and condition. Reject unapproved substitutions; check against released BOM.
2 Unpowered dry fit	Inspect cut edges, panel joints, feet, fan mount and guard fit. Keep electronics disconnected. Resolve tolerance clashes before sealing.
3 Seals and filters	Fit reviewed ledges, penetration seals, gaskets, stops and retainers in the documented order. Verify compression/flatness and supported service access.
4 Guards and wiring	Fit independently retained guards; install released harness restraints and enclosure. Verify cable isolation from moving parts and guarded openings.
5 Commission	Qualified electrical/mechanical inspection and approved isolation/stop/restart/thermal checks. Record sign-off before performance operation.
6 Measure	Record airflow and pressure at reviewed settings; power and noise; controlled room PM decay/background only under approved method. No improvised smoke, powder or gas challenge.
7 Decide	Compare to agreed targets, keep failed results, investigate leakage/low flow/noise. Update design and repeat relevant verification before expansion.

Blank CSVs are included. Instrument ranges, calibration and fixture loading must match the approved test method. Consumer PM readings do not certify filter class. Do not infer outdoor radius from an indoor decay test.

Parts funding and evidence handover

COMBINED_PARTS_REGISTER.csv separates exact OEM references, custom geometry and unresolved hardware/services. CAD_PART_INVENTORY.csv is a modeled-object list, not an order list. FASTENER_SCHEDULE.csv groups hardware; do not sum both as additional purchases.

Use staged funding: review and engineering closure; then released components/fabrication; then commissioning and calibrated testing. Request actual quotes and lead times against the approved revision. No INR total is asserted without quotations.

The ZIP includes selected current native/STEP/exploded files, 15 DXFs, current engineering evidence and standalone calculation inputs. No old CAD branches, downloaded manufacturer manuals, email address or conversation records are included. OEM facts must still be checked against delivered parts.

Sources are recorded in electrical_package/sources.json, harness_detail/README.md and GUARD_ACCESS_DECISION.md. Principal official sources include ARCTIC P14 Max support, Noctua NV-PS1/NA-FC1/NA-FH1 manuals and IKEA India STARKVIND104.633.30. Archived references are evidence, not current stock or price guarantees.

Current filter comparison: scripts/analysis/d01_current_filter_screen.py uses the current R03M budget. Supply per-filter m3/h and Pa with exact part/revision, source, CLEAN or LOADED condition and its conditioning reference. It refuses extrapolation and preserves input hashes. Positive model margin is conditional, not approval. Do not use the earlier R01 offer checker for this design. See TOOL_USAGE.md.

The geometric assembly is ours; proprietary purchased parts retain their manufacturers' rights. This package is for internal review and does not assert patentability or a license to redistribute OEM materials.

Handback requested from the team: one annotated review register and one coordinated revision, not parallel incompatible assemblies. Keep changes traceable to R03M and record any accepted change to safety requirements. Do not mark all remaining work complete while these decisions remain open.

Annexes follow: mechanical R03M; electrical E00; validation V00. All are reference snapshots and remain review-only. Corrections in this review section take precedence. The complete machine has not been built or tested.

Appendix A

Mechanical drawings

Reference snapshot for D01 R03M internal review

Read the current review section and TOOL_USAGE.md first.

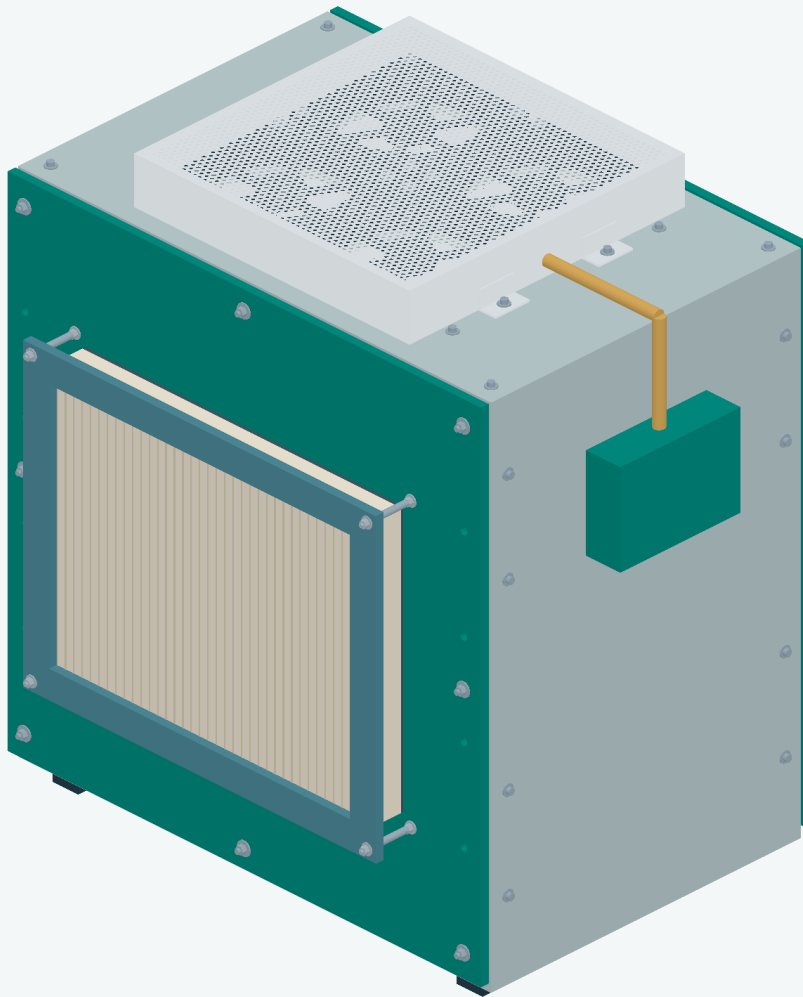
Current scripts and corrections supersede older analysis instructions.

Review drawings require release before cutting or wiring.

No physical performance or safety approval is recorded.

Source: AQI_D01_MECHANICAL_PACKAGE.pdf

One mechanical revision, not another design branch



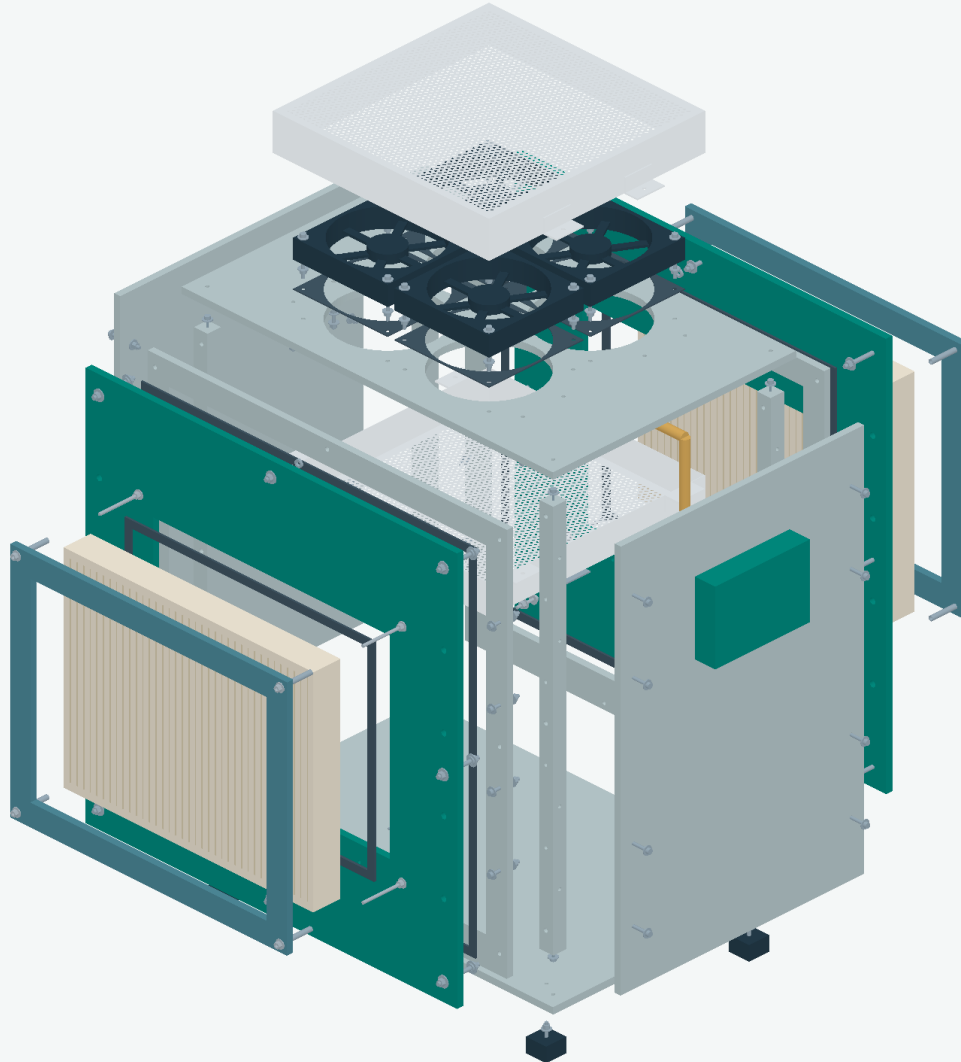
Same cabinet/filter arrangement. Modeled physical envelope 560 x504 x631 mm; excludes control/cable reservations.

Added: complete proposed fan/cabinet/M5 stacks, shorter rods, positive foot bolts, seal-gap stops, and joined guard subassemblies.

493 valid objects; 119,805 pair checks clear. Only reservation pairs excluded. STEP/native checks and both 200 mm filter-removal sweeps pass.

Still conditional: actual material and weld strength, filter/gasket fit, bearing/preload, stability, guarding tests and electrical enclosure interface. These are not physical PASS results.

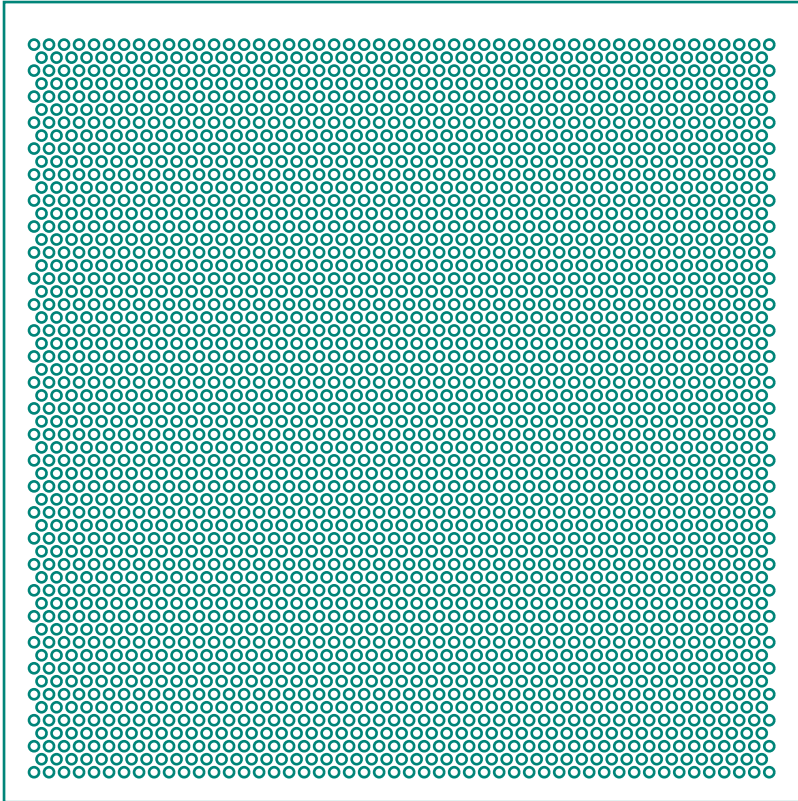
Exploded assembly / same modeled parts



Exploded positions explain grouping; they are not a verified removal trajectory. The complete assembly and exploded native files use the same part identities. Hardware remains schematic: no real threads or weld beads.

Assemble cabinet and reducer mountings before installing the fan plate/inner guard. Top centre M5 nuts are close to the internal guard: do not assume a socket fits after the guard is installed.

Fabrication: two joined guard assemblies



PROPOSED route: fabricated steel guards with joined face, four corner seams and welded mounting-angle laps. This replaces loose sheets/rivets; independent guard-to-plate bolts remain.

Blanks: face320 x320 x1 (quantity2); outer sides320 x50 x1 (2), ends318 x50 x1 (2); inner sides320 x40 x1 (2), ends318 x40 x1 (2). Inner skirt starts at z531, above its1 mm bottom face, eliminating double-counted stock overlap.

W1 joins the face perimeter. W2 joins four vertical corners. W3 joins each mounting-angle lap along its two short edges and top edge. These define joint locations, NOT a released weld size/process. Thin-sheet distortion, toe clearance and capacity require qualified fabrication review.

Proposed face pattern: 2822 holes,4 mm diameter,6 mm staggered pitch, at least10 mm solid border. Exact hole area 0.03546 m2 each.2D DXF supplied;3D face still an envelope. No finger-access approval.

No old rivet holes in the new blanks. Deburr all cut edges; no sharp projections or weld spatter near filters/fans. Material grade/coating and welding procedure are not yet selected.

W1: continuous face-to-skirt perimeter joint

Fastening schedule / installed stacks

M5 reducer studs / quantity 16

12 x60 mm;4 x40 mm;32 nuts8AFx5;32 washers15ODx1.2 (bore5.3 assumed). PROPOSED; external thread projection2.8mm; inner3.8mm

Seal compression limiters / quantity 24

16 outer OD10/ID5.5 x3;8 inner OD6.5/ID4.5 x36. ASSUMED nominal gaps; force/tolerance/actual filter thickness unknown

Fan fastenings / quantity 16

M4x50 cap screw;32 washers12ODx1;16 nuts7AFx5. PROPOSED;6mm nominal thread projection; fan corner bearing strength unknown

Cabinet through joints / quantity 32

M4x40 cap screws;64 washers12ODx1;32 nuts7AFx5. PROPOSED;4mm nominal thread projection; wood bearing/edge and clamp load not approved

Top/base wood screws / quantity 8

4 top4x20;4 base4x25;8 washers;head7dia x4 envelope. Core only; thread/pilot, wood grade and end-grain withdrawal must be verified; never lift by lid

Foot retention / quantity 4

M4x30 cap screw;8 washers12ODx1;4 nuts7AFx5;4 feet with12dia x9 recess. PROPOSED; head4mm above floor;3mm nominal thread projection; foot material strength unknown

Retained adapter M4 hardware / quantity 8

8 M4x80 studs;4 M4x25 ledge screws;28 nuts;32 washers. R02H geometry retained; sealing washers separate

Independent guard bolts / quantity 8

8 M4x20 screws;8 nuts;16 washers. R02G mounting stacks retained; welded cage replaces rivets

Seal routing and compression stops

Air path: room -> filter media -> sealed filter border -> reducer -> cabinet plenum -> fan gasket -> fans/outlet. Each bypass joint must be sealed; metal washers and nuts are not air seals.

Outer seal: existing 495.3 mm square outside / 475.3 mm inside ring, 3 mm installed. Sixteen proposed OD10/ID5.5 x 3 mm hard spacers sit outside this ring at the M5 locations. They limit the nominal plate gap; they do not select gasket force.

Filter seal: 370 x 290 mm outside / 350 x 270 mm inside, 3 mm installed. Eight proposed OD6.5/ID4.5 x 36 mm sleeves sit between fixed M4 nuts and retainers. With the current nut/washer stack, this gives 43 mm from reducer face to retainer: 40 mm filter + 3 mm gasket.

Tolerance warning: assuming a 4 mm free gasket, a 39/40/41 mm filter in that fixed gap implies 0/25/50% compression. These are illustrations, not measured tolerances. Actual gasket force, filter landing width and thickness must govern final spacer lengths. No tightening torque is invented.

Seal all fixed cabinet edge joints and mounting feedthroughs with compatible selected products. Keep the removable top seam serviceable; do not depend on uncharacterized glue as a structural joint. A continuous perimeter tape/seal route is proposed, product/adhesion/aging unresolved.

Fan seals remain 1 mm installed envelopes. Foot and new joint penetrations also need bypass inspection. Leak testing must demonstrate the whole assembled path; a CAD ring and ordinary PM readings do not certify filtration efficiency.

Supports, weight and stability: what is known

Filter weight sits on two proposed bent ledges per filter, separate from clamp friction. Each ledge is 30 mm wide, 43 mm shelf, 20 mm downstand, 2 mm nominal metal. Bend radius and load capacity remain unverified.

Feet now have positive through fastenings: 30 x 30 x 20 mm proposed blocks; 4.5 mm through hole, 12 mm diameter x 9 mm underside recess. M4x30 screw head is recessed 4 mm above floor; two washers and upper nut complete each joint. Actual foot material, tear/compression capacity and friction are unknown.

Assumed-material CAD subtotal: 12.19 kg. It excludes fans, filters, feet, seals and controls. It uses 650 kg/m³ panels, 600 cleats and 7850 metal; these are assumptions. Guard holes are not subtracted from 3D mass. This is NOT the device weight or its full centre of gravity.

Static tipping sensitivity assumes a centred whole machine, support polygon x25..525/y15..345 and a push at 631 mm height. At hypothetical 15 kg whole mass, front/back tipping balance is 38.5 N; at 20 kg, 51.3 N. No safety factor, sliding, impact or public misuse is included.

Consequences: supervised level indoor floor only after stability review; protect against cable pull and handling impacts. Do not lift by the lid, guards or filter retainers. No public/outdoor deployment or invented ballast/anchor selection.

Assembly sequence / one coherent build route

1. Review hold points before ordering/cutting: real filter and gasket dimensions, material grades, joint/weld method, fastener/tool access and protective enclosure interface.
2. Prepare the 10 panel blanks and revised guard blanks. The new base adds foot holes. Verify dimensions and hole locations against the current revision only; dry-fit for square and flatness.
3. Assemble base, side/seal panels and cleats with the proposed through-fastener stacks. Install foot bolts from the recessed underside. Do not crush wood or foot material; torque values await material/joint review.
4. Install outer gasket, 3 mm stops, reducers and short M5 rods. Hold inside nuts before the fan plate/inner guard blocks access. Seal fixed joints and feedthroughs using verified products.
5. Install ledges, M4 studs/sealing washers/fixed nuts, then verified filter gasket and filter. Fit measured-length stop sleeves, retainers, washers and service nuts evenly. Confirm actual compression, no frame damage and support contact.
6. Fabricate guards using reviewed joints; independently mount the inner guard to the fan plate. Install fan gaskets and 16 fan bolt stacks, then outer guard using its separate four mounts. Ensure clearance after fastening.
7. Fit the top and complete the reviewed control-enclosure/cable interface from package 2. Keep cables clear of impellers, service openings and sharp edges. Do not energize at this stage.
8. Perform mechanical dry-fit, guard/access, seal, load and stability checks; then qualified electrical commissioning. Physical airflow/filter trials belong to package 3.

Release record / exact remaining holds

DIGITAL CHECKED: assembly shapes, modeled interference, native/STEP export, matching panel faces, proposed full fastener stacks, both filter extraction sweeps, and reproducible mass/tolerance arithmetic. These do not prove physical integrity.

HOLD M1 - Materials and joints: actual panel/cleat/foot/metal properties; wood bearing and end-grain withdrawal, panel/retainer bending, bracket capacity, weld procedure/quality and guard deflection. No load rating is released.

HOLD M2 - Seals and fit: actual filter border/thickness/mass, selected gasket force/deflection and products, final stop lengths, foot and new fastener penetration sealing, filter frame loads and physical leak results.

HOLD M3 - Access and stability: actual tool/hand paths, guard reach/probe tests, loose/captive hardware risk, full weight/CG and tipping/sliding/handling checks. Power must be isolated before access; no running-fan demonstration authorized.

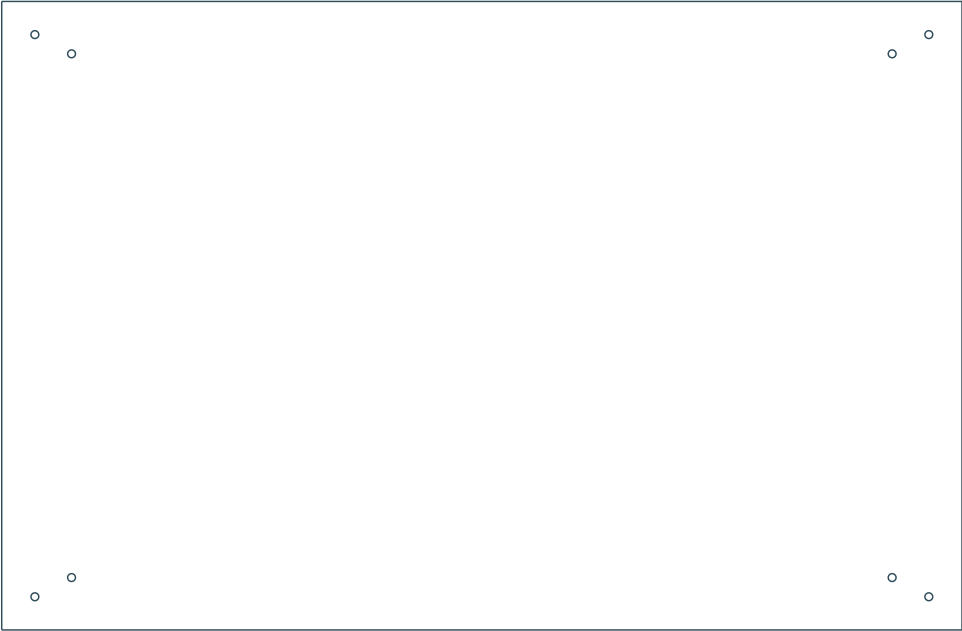
INTERFACE TO PACKAGE2: exact control enclosure, cable glands/restraints and protective isolation/restart system are not specified by this mechanical package. The present exterior box/cable CAD remains a reservation, not a manufactured component.

No purchase, supplier contact, software installation, fabrication or physical test occurred. This is one mechanical review package with substantial detailing completed, NOT a claim that every engineering input or construction release is complete.

M01_Base / current panel geometry

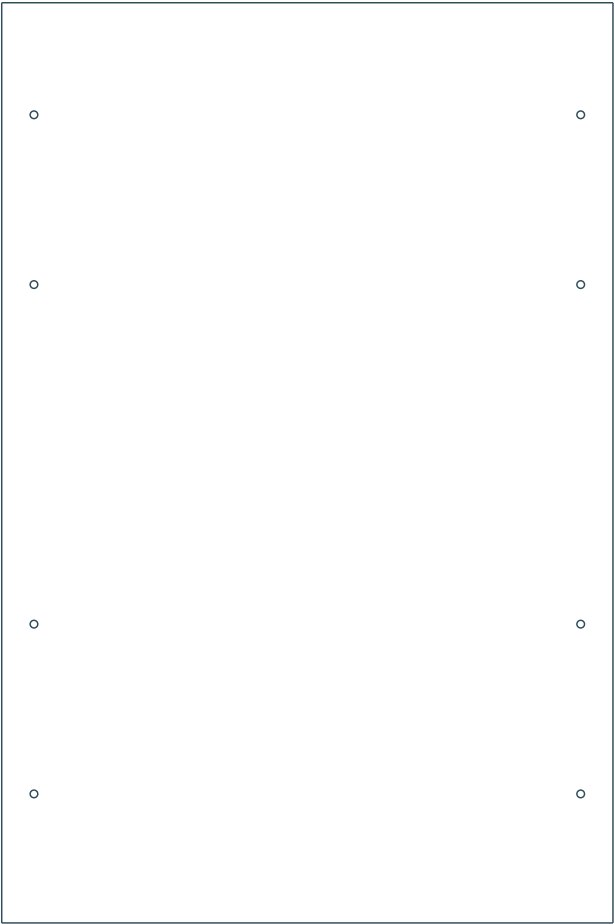
Local xy axes; lower-left datum. Circle list: (x,y), diameter in mm.
Same-name DXF supplied. Geometry only, not toleranced cutting approval.

- | | |
|--------------------------|---------------------------|
| (19.00, 19.00) dia 4.50 | (19.00, 341.00) dia 4.50 |
| (40.00, 30.00) dia 4.50 | (40.00, 330.00) dia 4.50 |
| (510.00, 30.00) dia 4.50 | (510.00, 330.00) dia 4.50 |
| (531.00, 19.00) dia 4.50 | (531.00, 341.00) dia 4.50 |



550.00 x 360.00 x 9.00 mm / qty1

M02_Left / current panel geometry

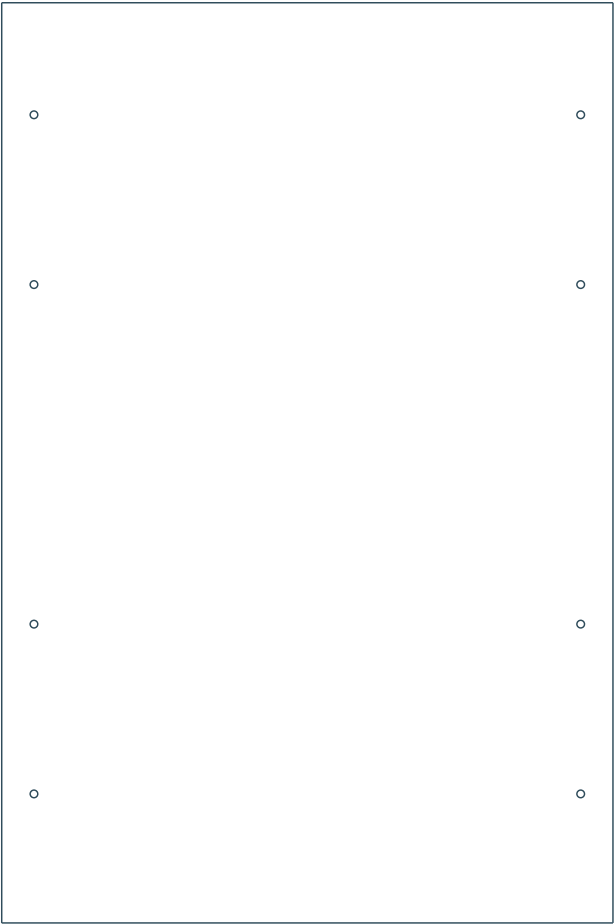


360.00 x 542.00 x 9.00 mm / qty1

Local yz axes; lower-left datum. Circle list: (x,y), diameter in mm.
Same-name DXF supplied. Geometry only, not toleranced cutting approval.

(19.00, 76.00) dia 4.50	(19.00, 176.00) dia 4.50
(19.00, 376.00) dia 4.50	(19.00, 476.00) dia 4.50
(341.00, 76.00) dia 4.50	(341.00, 176.00) dia 4.50
(341.00, 376.00) dia 4.50	(341.00, 476.00) dia 4.50

M02_Right / current panel geometry



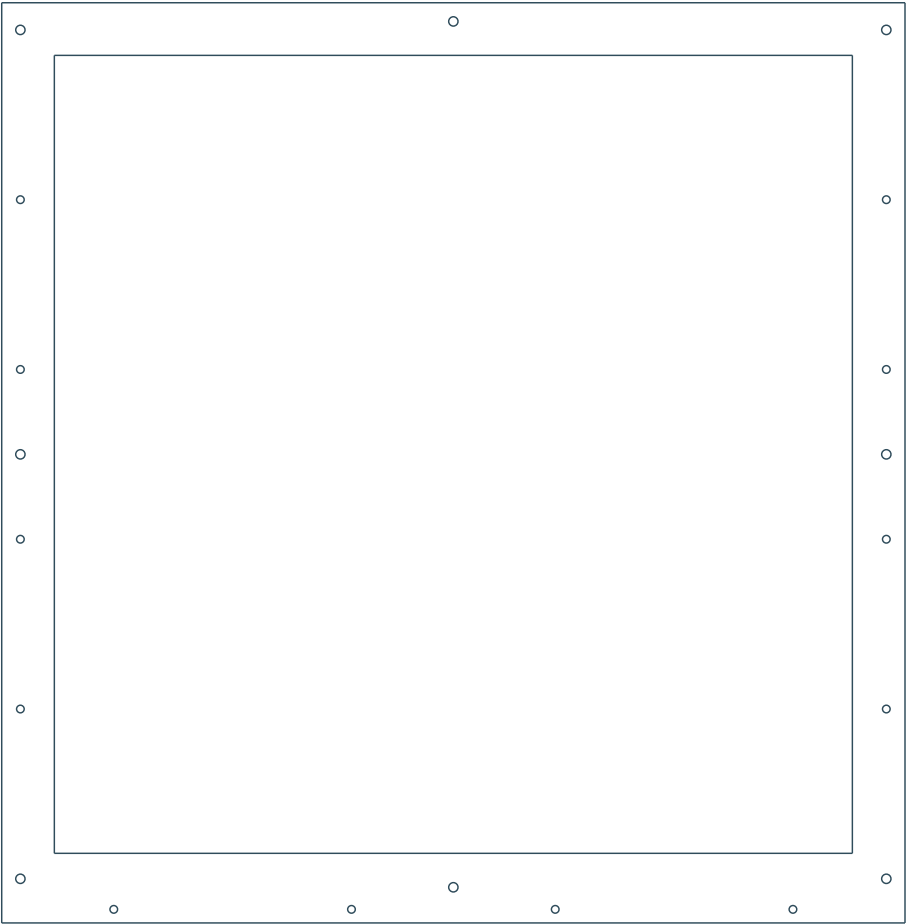
360.00 x 542.00 x 9.00 mm / qty1

Local yz axes; lower-left datum. Circle list: (x,y), diameter in mm.
Same-name DXF supplied. Geometry only, not toleranced cutting approval.

(19.00, 76.00) dia 4.50	(19.00, 176.00) dia 4.50
(19.00, 376.00) dia 4.50	(19.00, 476.00) dia 4.50
(341.00, 76.00) dia 4.50	(341.00, 176.00) dia 4.50
(341.00, 376.00) dia 4.50	(341.00, 476.00) dia 4.50

M03_Seat_Front / current panel geometry

Local xz axes; lower-left datum. Circle list: (x,y), diameter in mm.
Same-name DXF supplied. Geometry only, not toleranced cutting approval.

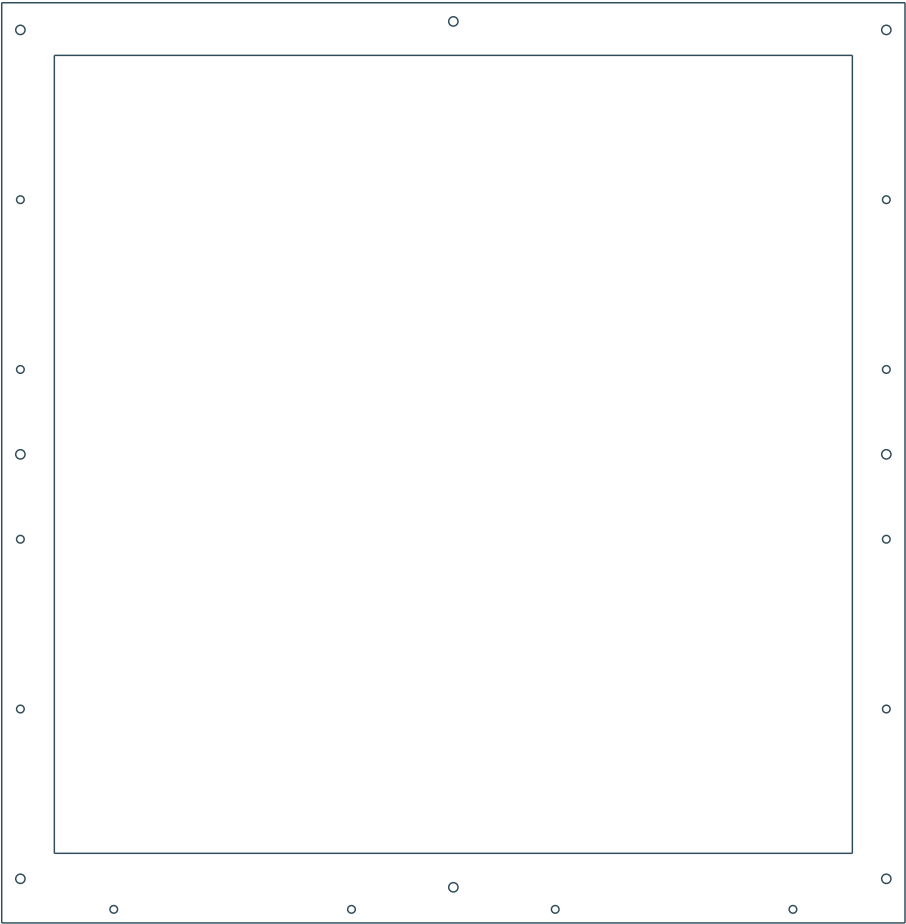


532.00 x 542.00 x 9.00 mm / qty1

- | | |
|---------------------------|---------------------------|
| (11.00, 126.00) dia 4.50 | (11.00, 226.00) dia 4.50 |
| (11.00, 326.00) dia 4.50 | (11.00, 426.00) dia 4.50 |
| (66.00, 8.00) dia 4.50 | (206.00, 8.00) dia 4.50 |
| (326.00, 8.00) dia 4.50 | (466.00, 8.00) dia 4.50 |
| (521.00, 126.00) dia 4.50 | (521.00, 226.00) dia 4.50 |
| (521.00, 326.00) dia 4.50 | (521.00, 426.00) dia 4.50 |
| (11.00, 26.00) dia 5.50 | (11.00, 276.00) dia 5.50 |
| (11.00, 526.00) dia 5.50 | (266.00, 21.00) dia 5.50 |
| (266.00, 531.00) dia 5.50 | (521.00, 26.00) dia 5.50 |
| (521.00, 276.00) dia 5.50 | (521.00, 526.00) dia 5.50 |

M03_Seat_Rear / current panel geometry

Local xz axes; lower-left datum. Circle list: (x,y), diameter in mm.
Same-name DXF supplied. Geometry only, not toleranced cutting approval.

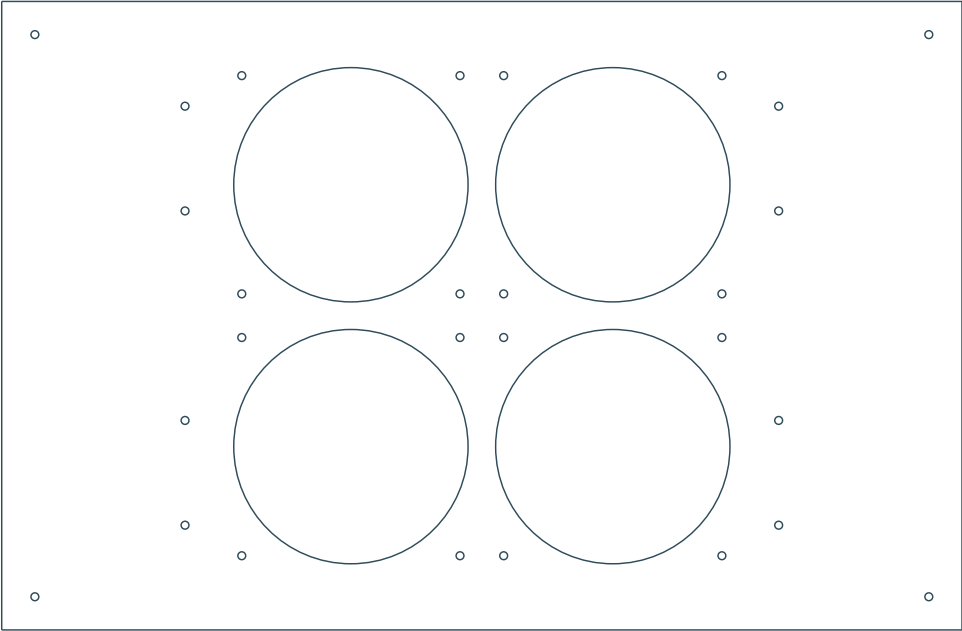


532.00 x 542.00 x 9.00 mm / qty1

- | | |
|---------------------------|---------------------------|
| (11.00, 126.00) dia 4.50 | (11.00, 226.00) dia 4.50 |
| (11.00, 326.00) dia 4.50 | (11.00, 426.00) dia 4.50 |
| (66.00, 8.00) dia 4.50 | (206.00, 8.00) dia 4.50 |
| (326.00, 8.00) dia 4.50 | (466.00, 8.00) dia 4.50 |
| (521.00, 126.00) dia 4.50 | (521.00, 226.00) dia 4.50 |
| (521.00, 326.00) dia 4.50 | (521.00, 426.00) dia 4.50 |
| (11.00, 26.00) dia 5.50 | (11.00, 276.00) dia 5.50 |
| (11.00, 526.00) dia 5.50 | (266.00, 21.00) dia 5.50 |
| (266.00, 531.00) dia 5.50 | (521.00, 26.00) dia 5.50 |
| (521.00, 276.00) dia 5.50 | (521.00, 526.00) dia 5.50 |

M06_Fan_plate / current panel geometry

Local xy axes; lower-left datum. Circle list: (x,y), diameter in mm.
Same-name DXF supplied. Geometry only, not toleranced cutting approval.

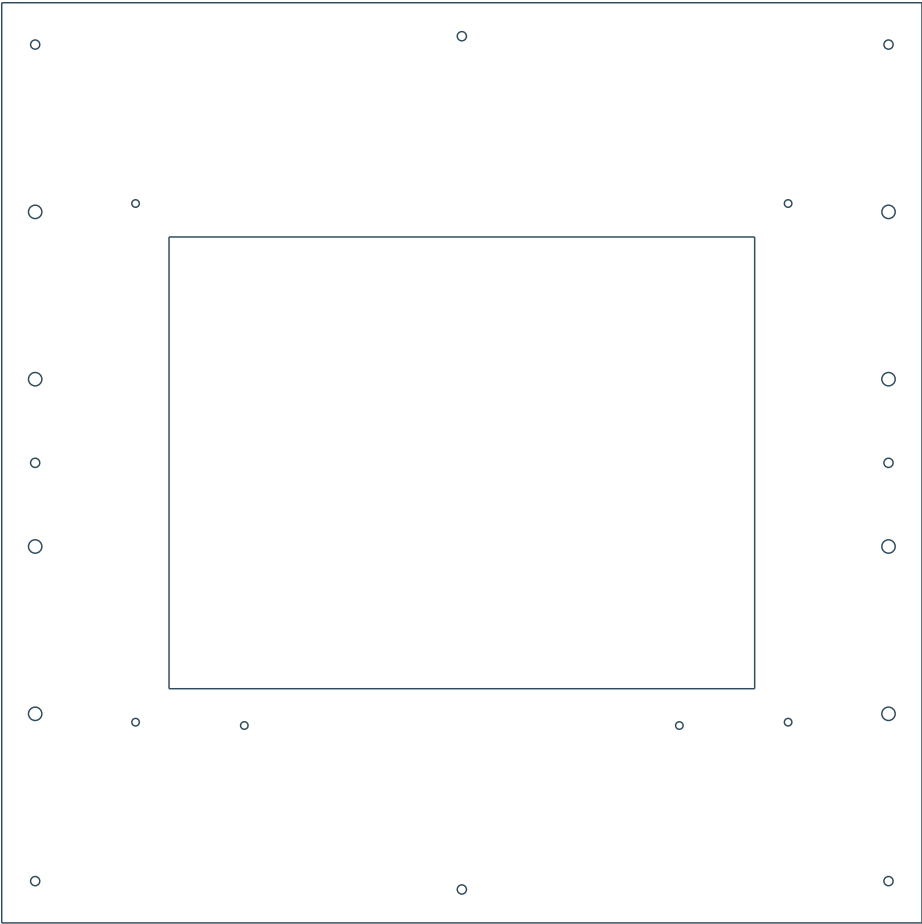


550.00 x 360.00 x 9.00 mm / qty1

(19.00, 19.00) dia 4.50	(19.00, 341.00) dia 4.50
(105.00, 60.00) dia 4.50	(105.00, 120.00) dia 4.50
(105.00, 240.00) dia 4.50	(105.00, 300.00) dia 4.50
(137.50, 42.50) dia 4.50	(137.50, 167.50) dia 4.50
(137.50, 192.50) dia 4.50	(137.50, 317.50) dia 4.50
(262.50, 42.50) dia 4.50	(262.50, 167.50) dia 4.50
(262.50, 192.50) dia 4.50	(262.50, 317.50) dia 4.50
(287.50, 42.50) dia 4.50	(287.50, 167.50) dia 4.50
(287.50, 192.50) dia 4.50	(287.50, 317.50) dia 4.50
(412.50, 42.50) dia 4.50	(412.50, 167.50) dia 4.50
(412.50, 192.50) dia 4.50	(412.50, 317.50) dia 4.50
(445.00, 60.00) dia 4.50	(445.00, 120.00) dia 4.50
(445.00, 240.00) dia 4.50	(445.00, 300.00) dia 4.50
(531.00, 19.00) dia 4.50	(531.00, 341.00) dia 4.50
(200.00, 105.00) dia 134.20	(200.00, 255.00) dia 134.20
(350.00, 105.00) dia 134.20	(350.00, 255.00) dia 134.20

A01_Reducer_Front / current panel geometry

Local xz axes; lower-left datum. Circle list: (x,y), diameter in mm.
Same-name DXF supplied. Geometry only, not toleranced cutting approval.



- | | |
|---------------------------|---------------------------|
| (80.00, 120.00) dia 4.50 | (80.00, 430.00) dia 4.50 |
| (145.00, 118.00) dia 4.50 | (405.00, 118.00) dia 4.50 |
| (470.00, 120.00) dia 4.50 | (470.00, 430.00) dia 4.50 |
| (20.00, 25.00) dia 5.50 | (20.00, 275.00) dia 5.50 |
| (20.00, 525.00) dia 5.50 | (275.00, 20.00) dia 5.50 |
| (275.00, 530.00) dia 5.50 | (530.00, 25.00) dia 5.50 |
| (530.00, 275.00) dia 5.50 | (530.00, 525.00) dia 5.50 |
| (20.00, 125.00) dia 8.00 | (20.00, 225.00) dia 8.00 |
| (20.00, 325.00) dia 8.00 | (20.00, 425.00) dia 8.00 |
| (530.00, 125.00) dia 8.00 | (530.00, 225.00) dia 8.00 |
| (530.00, 325.00) dia 8.00 | (530.00, 425.00) dia 8.00 |

550.00 x 550.00 x 9.00 mm / qty1

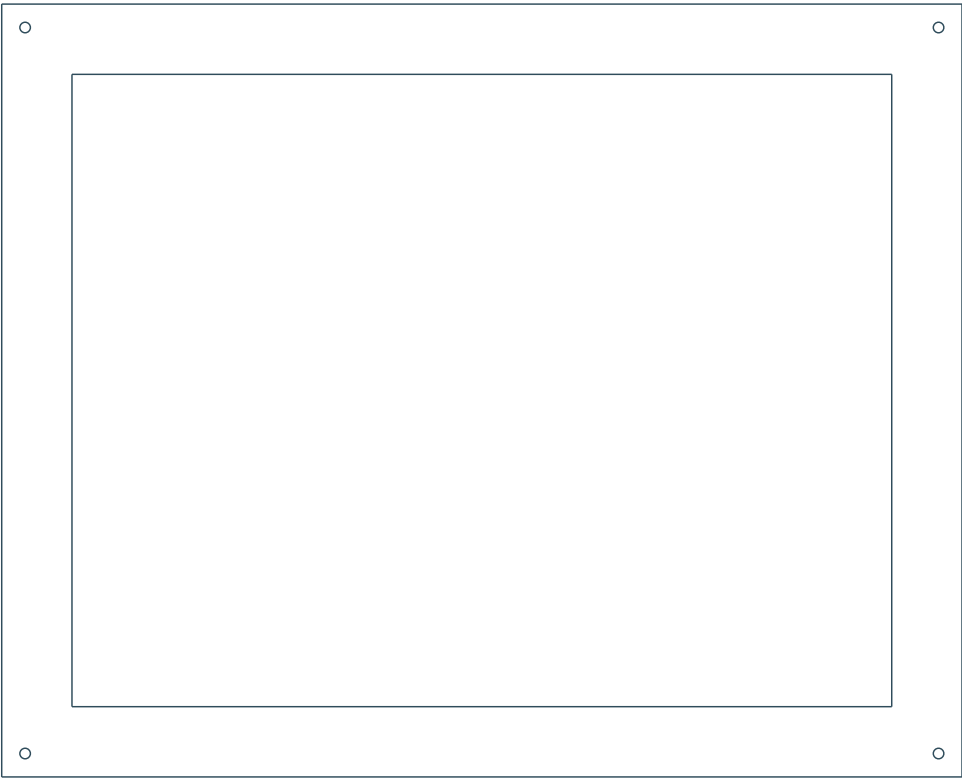
A04_Retainer_Front / current panel geometry

Local xz axes; lower-left datum. Circle list: (x,y), diameter in mm.
Same-name DXF supplied. Geometry only, not toleranced cutting approval.

- (10.00, 10.00) dia 4.50

(400.00, 10.00) dia 4.50
- (10.00, 320.00) dia 4.50

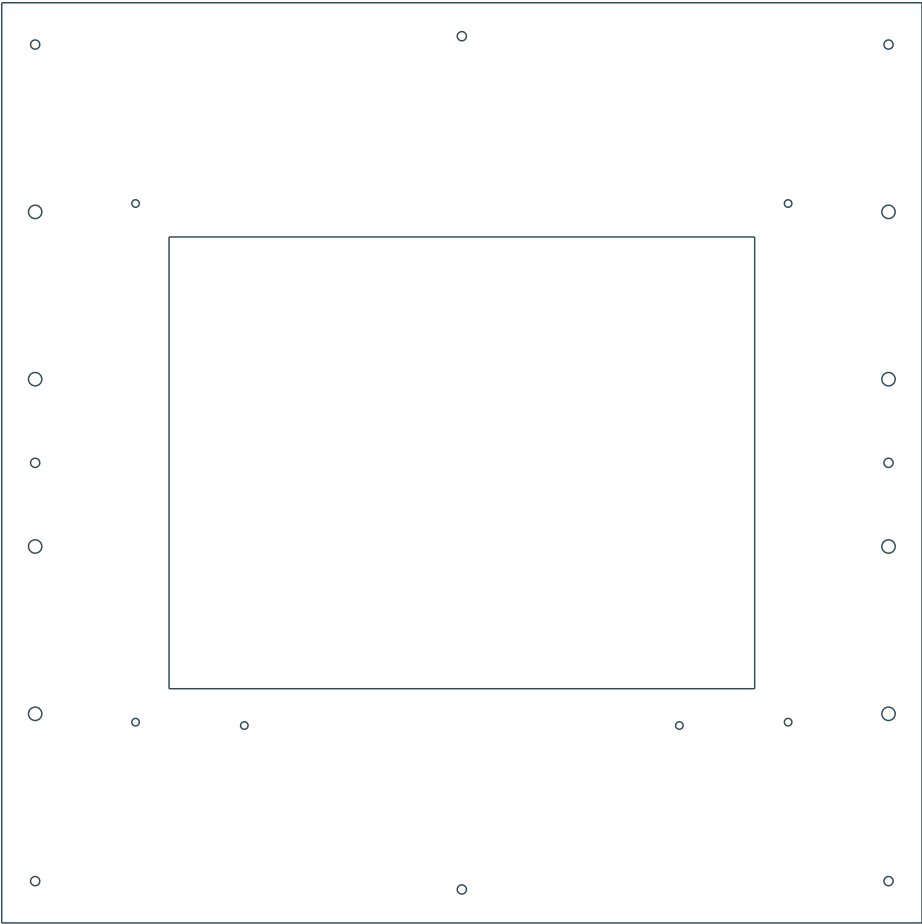
(400.00, 320.00) dia 4.50



410.00 x 330.00 x 9.00 mm / qty1

A01_Reducer_Rear / current panel geometry

Local xz axes; lower-left datum. Circle list: (x,y), diameter in mm.
Same-name DXF supplied. Geometry only, not toleranced cutting approval.



- (80.00, 120.00) dia 4.50
- (145.00, 118.00) dia 4.50
- (470.00, 120.00) dia 4.50
- (20.00, 25.00) dia 5.50
- (20.00, 525.00) dia 5.50
- (275.00, 530.00) dia 5.50
- (530.00, 275.00) dia 5.50
- (20.00, 125.00) dia 8.00
- (20.00, 325.00) dia 8.00
- (530.00, 125.00) dia 8.00
- (530.00, 325.00) dia 8.00

- (80.00, 430.00) dia 4.50
- (405.00, 118.00) dia 4.50
- (470.00, 430.00) dia 4.50
- (20.00, 275.00) dia 5.50
- (275.00, 20.00) dia 5.50
- (530.00, 25.00) dia 5.50
- (530.00, 525.00) dia 5.50
- (20.00, 225.00) dia 8.00
- (20.00, 425.00) dia 8.00
- (530.00, 225.00) dia 8.00
- (530.00, 425.00) dia 8.00

550.00 x 550.00 x 9.00 mm / qty1

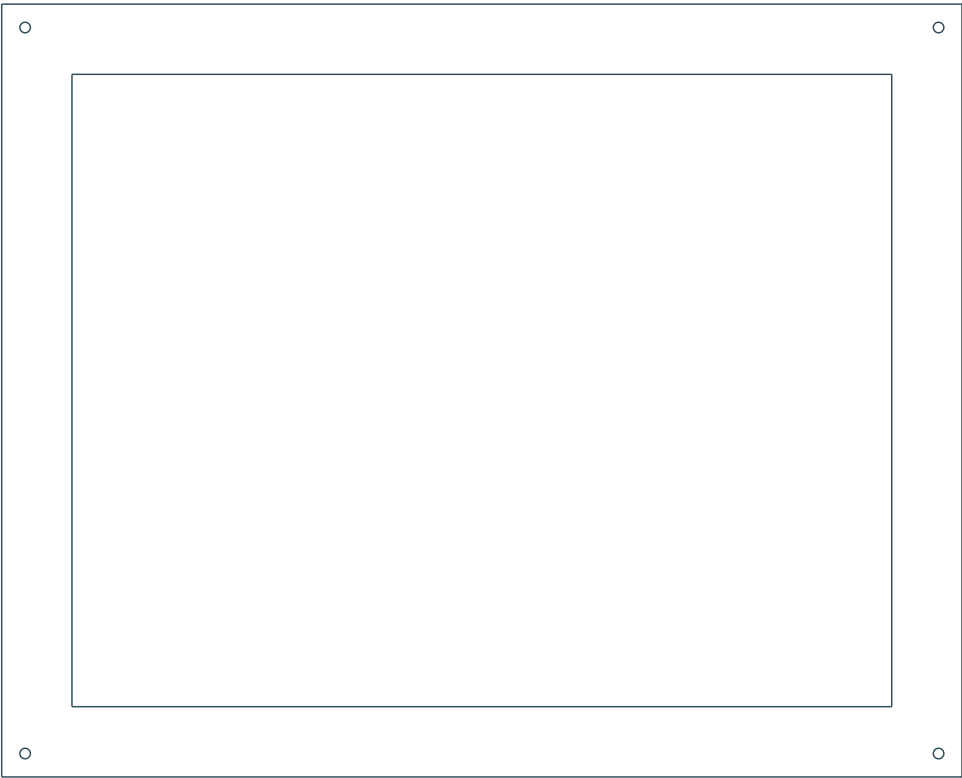
A04_Retainer_Rear / current panel geometry

Local xz axes; lower-left datum. Circle list: (x,y), diameter in mm.
Same-name DXF supplied. Geometry only, not toleranced cutting approval.

- (10.00, 10.00) dia 4.50

(400.00, 10.00) dia 4.50
- (10.00, 320.00) dia 4.50

(400.00, 320.00) dia 4.50



410.00 x 330.00 x 9.00 mm / qty1

Appendix B

Electrical review

Reference snapshot for D01 R03M internal review

Read the current review section and TOOL_USAGE.md first.

Current scripts and corrections supersede older analysis instructions.

Review drawings require release before cutting or wiring.

No physical performance or safety approval is recorded.

Source: AQI_D01_ELECTRICAL_REVIEW.pdf

Powering the first prototype

Package 2: one electrical review for the existing R03M assembly

Decision: retain four 12 V fans and an external OEM power adapter as the reference route. Keep mains wiring outside the cabinet. Use local speed adjustment; no app or custom power PCB is needed for the first experiment.

What this closes: the OEM connection reference, nominal load calculation, named interfaces, enclosure fit screen, restart acceptance model and fault-test specification are now in one reproducible package.

What it does not close: the rated protective stop/restart circuit. The OEM chain can run fans, but does not itself satisfy our manual-restart requirement. It is not an approved wiring diagram.

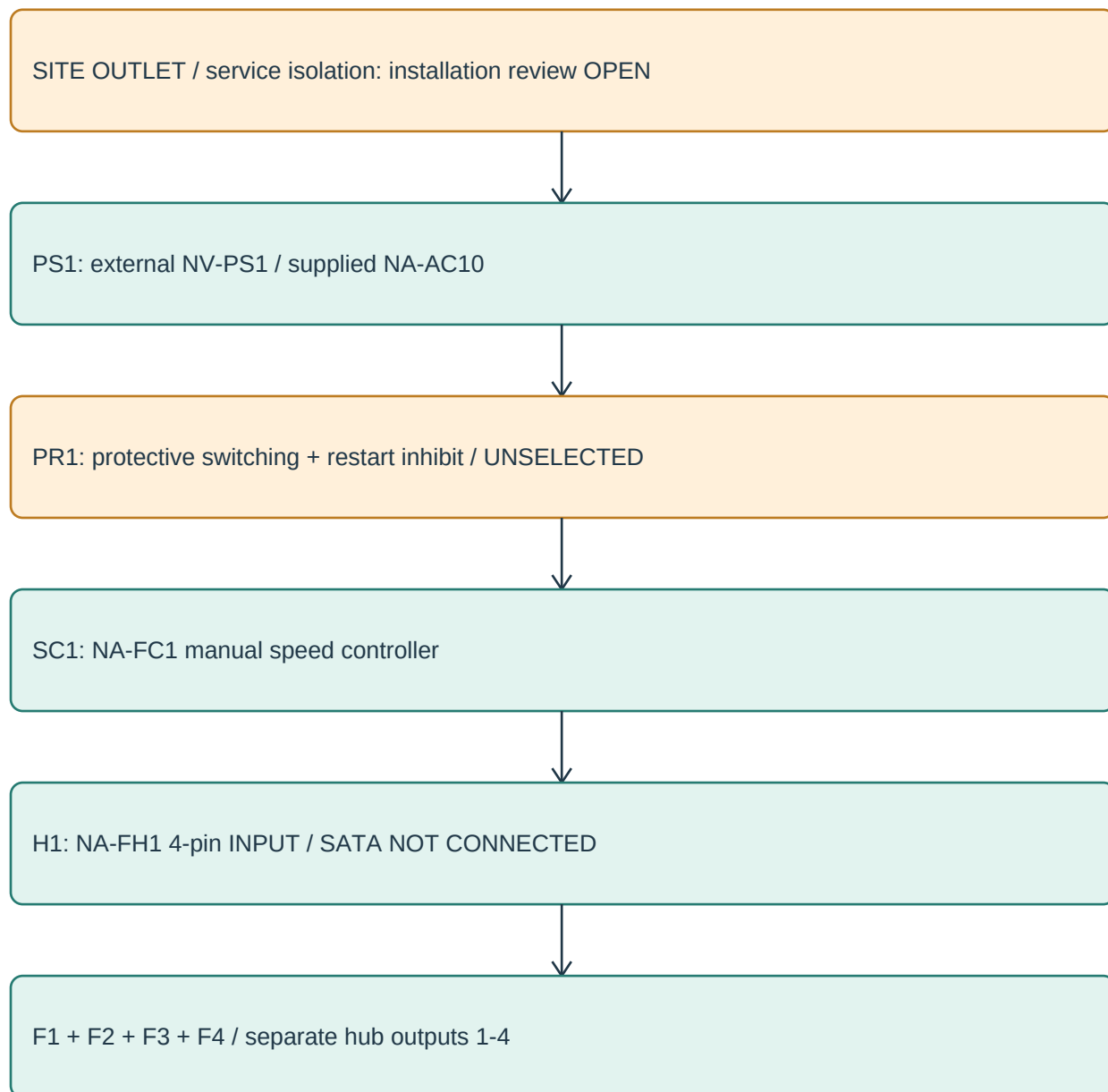
DETECTED: four fans total 1.40 A / 16.8 W nominal. The adapter and selected hub input are limited to 2 A / 24 W. The difference is 0.60 A before controller/hub consumption, startup and tolerances; it is not proven spare capacity.

RECOMMENDATION: retain the external adapter, fixed guards and separate protective module approach. Do not add a 24 V safety relay merely to fill a BOM: a separate control supply can remain alive while the 12 V supply trips and recovers.

Physical prototypes: 0. No components purchased, no electrical tests performed. This indoor demonstrator does not prove outdoor cleaning radius, heat tolerance or vandal resistance.

E01 / power connection reference

Block-level architecture; lines are interfaces, not conductor or terminal instructions



PR1 is a **proposed insertion boundary**, not a verified inline product or terminal assignment. Its sensing, switching, fault response and connection harness are still unselected. The unmodified OEM reference chain omits PR1 and therefore is NOT our released system.

Logger/instruments remain electrically separate. No SATA, motherboard, USB fan-power source or remote-start path is part of this reference.

E05 / load and rating screen

Calculated from published nominal values; no startup waveform or physical test

Fan load: $4 \times 0.35 \text{ A} = 1.40 \text{ A}$; $12 \text{ V} \times 1.40 \text{ A} = 16.8 \text{ W}$. NA-FC1 rating is 3 A / 36 W, but it does not raise the 24 W supply/hub-path ceiling. [S1-S4]

Startup sensitivity, not predicted inrush:

1.00 x nominal: 1.40 A; difference to 2 A = +0.60 A.

1.25 x: 1.75 A; difference = +0.25 A.

1.50 x: 2.10 A; difference = -0.10 A.

2.00 x: 2.80 A; difference = -0.80 A.

All exclude auxiliary current. Startup approval remains OPEN.

Cable calculation example only: with assumed copper resistivity 0.0175 ohm mm²/m, 1 m one-way length, 0.5 mm² conductors and 1.4 A, loop drop is 0.098 V before connectors. This does NOT select cable size: actual length, insulation, temperature, fault protection and terminations remain unknown.

Protection is not sized from 1.4 A alone. Supply short-circuit behavior, switching DC capability, contact welding, branch cable withstand and protection coordination must be established together. No invented fuse, breaker or wire rating is issued.

Temperature: NV-PS1 is specified 0-40 C; the fan has the same project ceiling. Ambient rating is not allowance for a hotter closed enclosure. Keep the adapter external; verify actual worst-case internal temperatures. No India outdoor approval follows.

Class II adapter: do not add an improvised earth connection to it. Whole-assembly bonding and installation protection need review; its component class does not certify the cabinet. Plug types C/A/G supplied are not proof of suitability for the actual Indian outlet. [S2]

E02 / restart and failure behavior

An executable requirements model is supplied - NOT controller firmware

Required sequence: power restored -> OFF; release START -> fresh START -> RUN REQUESTED. Fault -> TRIPPED; restore verified permissives -> release and press RESET -> OFF; release and press START -> RUN REQUESTED. STOP always wins.

Important finding 1: NA-FH1 uses self-resetting input protection. Its manual instructs disconnecting all power for one minute to reset after a trip. That is not a latched protective-stop guarantee. Recovery must not create an unintended run condition. [S5]

Important finding 2: the upstream RPM signal comes from output 1 only. Four status LEDs are not four independent protective feedback channels. One healthy fan cannot prove all four are operating. [S5/S7]

Important finding 3: a separate 24 V control supply may stay powered when the 12 V fan supply fails. A held run contact could then restart fans on recovery. Monitor the relevant power domain; also resolve downstream hub/fan recovery, which upstream voltage monitoring alone cannot detect.

Model checks: 14 named sequences and 1024 one-step combinations pass. Missing permissive, held START/RESET, power restoration, STOP priority and trip reset are covered. Ideal inputs only: no sensor, contact, timer, diagnostic coverage or safety integrity is proven.

Selection hold: no complete rated PR1 architecture has been established. A basic hold-in relay, PWM zero setting, hub fuse, fan RPM LED or Python test cannot close this gate. Determine required protective reliability and access/run-down measures before assigning terminal numbers.

E03 / E04: cabinet interfaces

Existing mechanical CAD is preserved; no unverified holes or enclosure added

OEM dimensions: hub 93 x 43 x 12.5 mm; controller 48 x 25 x 21 mm. The CAD reservation is 140 x 100 x 40 mm, not selected internal enclosure space. Controller + hub can only be screened geometrically; connectors, bend radii, cover, retention and thermal clearances are not included. [S3/S4]

Placement recommendation: keep only low-voltage OEM speed/hub parts in the side enclosure. Keep adapter and the eventual protective assembly external. Do not assume a relay panel fits in the reservation or hang its mass on an unreviewed cabinet joint.

Mechanical handoff: reservation x=550..590, y=110..250, z=390..490 mm. Reserved cable corridor runs (435,180,595) to (565,180,595) to (565,180,490). Coordinates use R03M CAD axes; this is space allocation, not a drilled feed-through.

Guard crossing remains OPEN: actual entry needs retention, edge protection and a reach-safe closure. Do not drill a generic hole in the guard. Obtain routed lengths and connector envelopes before choosing extensions. The included hub input cable is 300 mm; the reserved external route alone uses about 235 mm. [S7]

Interface schedule: J01 site supply; J02 OEM barrel adapter; J03 controller input; J04 hub input; J05-08 four fan leads; J09 protective permission; J10 independent logger; J11 service isolation. The accompanying CSV records each endpoint and its remaining limitation. No pin numbers inferred from cable colors.

Parts status: four named OEM products are reference candidates, not procurement release. Protective module, enclosure, switches, extensions, strain relief and isolation are unresolved. No prices or India stock invented; see BOM.csv.

E06 / physical acceptance and release

All physical cases below are NOT EXECUTED

Before power: confirm delivered product/manual revisions and source suitability; inspect guards, cable retention, polarity and terminations; complete applicable insulation, bonding and protection tests using qualified procedures. Do not improvise test voltages from this pack.

Controlled operating tests: no motion on power application; deliberate start/stop; held START during restoration; simultaneous START/STOP; RESET without START; held RESET on fault recovery; interruption of fan power with control power remaining. Record actual run-down before considering any access release.

Fault review / safe simulation: resolve hub self-reset, individual fan fault/recovery, missing PWM command, loss of protective input and switched-contact failure. Do not short live circuits, block blades, defeat guards or damage fans to simulate faults. Approved methods may use documents and isolated test fixtures.

Measurements: capture four-fan startup and steady current, controller/hub auxiliary load, fan-terminal voltage, supply recovery and enclosure temperature at the intended environment. RPM/PM readings do not certify safe standstill or filter sealing.

Finite release blockers:

1. Rated PR1 protective implementation including downstream recovery and access/run-down.
 2. Cable/enclosure/guard-entry detail and product/interface compatibility.
 3. Site supply review, applicable safety classification and qualified commissioning evidence.
- Mechanical strength/seal/guard holds and actual filter/airflow evidence remain separate gates.

Critical path: protective design and interface closure -> qualified design review -> authorized purchasing/fabrication -> inspected assembly -> controlled commissioning -> indoor airflow/particle test. Supplier delivery, review availability and physical results are unknown, so no completion date is invented.

Evidence and reproducibility

Sources checked 4 October 2026; manufacturer content remains owned by its publisher

S1 - ARCTIC P14 Max black

<https://www.arctic.de/en/P14-Max-Black/ACFAN00287A>

S2 - NV-PS1 specifications

<https://www.noctua.at/en/products/nv-ps1/specifications>

S3 - NA-FC1 specifications

<https://www.noctua.at/en/products/na-fc1/specifications>

S4 - NA-FH1 specifications

<https://www.noctua.at/en/products/na-fh1/specifications>

S5 - NA-FH1 manual

https://cdn.noctua.at/media/noctua_na_fh1_manual_spot_en.pdf?download=true

S6 - NV-PS1 manual

https://cdn.noctua.at/media/noctua_nv_ps1_manual_en_web.pdf?download=true

S7 - NA-FH1 features

<https://www.noctua.at/en/products/na-fh1/features>

Run scripts/analysis/d01_control_acceptance.py for the abstract checks. Run scripts/reports/build_d01_electrical_package.py to reproduce calculations, schedules, report and bundle. Checks do not certify hardware. Historical 230 V fan wiring is not applicable to D01.

Appendix C

Validation procedure

Reference snapshot for D01 R03M internal review

Read the current review section and TOOL_USAGE.md first.

Current scripts and corrections supersede older analysis instructions.

Review drawings require release before cutting or wiring.

No physical performance or safety approval is recorded.

Source: AQI_D01_VALIDATION_PACKAGE.pdf

What the prototype must prove

First milestone: one safely assembled indoor unit with recorded airflow, filter pressure loss, electrical power, temperature and repeatable particle trends. Not a full outdoor tower and not a certified HEPA air cleaner.

This delivery: current R03M pressure screen, blank measurement workbook (three CSV sheets), a strict airflow calculator, synthetic regression tests and one physical test sequence. No fabricated results fill the empty fields.

Important correction: both guard patterns now use 0.0354623 m² open area each, not the earlier 0.036 m². With the same assumed loss coefficient, each guard loss increases by about 3.06%. This does not verify the loss coefficient or actual guard resistance.

Our project: a custom cabinet, seals, service arrangement, guarded four-fan assembly and measured system integration. Buying OEM fans/filters is normal; it does not make the system a proven product or a patentable invention.

Stop condition: do not energize the review-only assembly to collect these data. Mechanical strength/guard/seal review and the rated protective electrical design remain open. The test workbook is ready; the hardware is not yet released.

Evidence labels: OEM = published component data; ASSUMED = modeling choice; CALCULATED = arithmetic from stated inputs; MEASURED = physical record with method/instrument evidence. At present, physical prototypes and physical test runs both equal zero.

Updated pressure budget

Calculated, not measured: four fans in parallel add flow, not pressure. Two filter paths each carry half the total flow only under the equal-sharing assumption. Filter pressures must not be added as if the two paths were in series.

Central assumed reducer K = 1:

Total 200 m³/h / each filter 100: fan 32.60 Pa; other losses 5.92 Pa; filter allowance **26.68 Pa**.

Total 250 m³/h / each filter 125: fan 30.09 Pa; other losses 9.24 Pa; filter allowance **20.85 Pa**.

Total 300 m³/h / each filter 150: fan 27.63 Pa; other losses 13.31 Pa; filter allowance **14.32 Pa**.

Total 350 m³/h / each filter 175: fan 25.27 Pa; other losses 18.12 Pa; filter allowance **7.15 Pa**.

Total 400 m³/h / each filter 200: fan 22.83 Pa; other losses 23.66 Pa; filter allowance **-0.83 Pa**.

Meaning: the filter must fit inside the remaining pressure allowance at that same flow in this model. Negative allowance means the assumed non-filter losses already exceed the curve pressure. Positive allowance is NOT proof that the actual filter fits.

Model assumptions: OEM numeric curve at full 2800 rpm; density 1.2 kg/m³; 134.2 mm fan apertures; 350 x 270 mm opening per filter. Guard K1.5 each, plenum K1.5 and installation K1 on their stated velocity references. Reducer K0.5/1/2 sensitivities are included in the CSV.

Still unknown: actual installed fan curve, clean/loaded filter curve, guard K, bypass, uneven flow sharing and outlet recirculation. Reducer and plenum losses may overlap. Static pressure compatibility remains unresolved. These are not guaranteed bounds or a new CFD validation.

Decision: keep 300 m³/h as a provisional investigation point, not a product claim. Do not increase the power supply or fan speed beyond OEM limits to force that result. Change the design if actual flow/pressure evidence demands it.

Measure flow before claiming cleaning

1. Identify the exact assembly. Record CAD revision, delivered fan/filter identities, seals, guards, speed setting, room conditions and instrument/calibration references. Photograph the complete guarded configuration. Keep raw records unchanged.

2. Use a reviewed flow method. Prefer a suitable calibrated capture arrangement or an engineered duct traverse. A handheld reading at one outlet point multiplied by gross area is not a valid total flow measurement. Swirl, nonuniform velocity and both guard layers matter.

Fixture warning: a hood or duct can restrict this low-pressure fan system. Record fixture resistance and use a method validated for the installed configuration; do not call a loaded-fixture flow the unrestricted device flow without justification. Fixture dimensions and correction are not yet selected.

3. Record area-weighted cells. AIRFLOW_BLANK.csv contains empty placeholders, not an approved nine-point traverse. For an accepted normal-velocity map, $Q = 3600 \times \sum(\text{cell area} \times \text{mean normal velocity})$. Cells must cover the chosen plane once without overlap; the reviewer determines sampling and averaging.

4. Quantify error. The calculator reports only a conservative velocity-error contribution, $3600 \times \sum(\text{area} \times \text{velocity error bound})$. Area, temporal drift, probe alignment, fixture loading/leakage and spatial sampling require a separate full uncertainty budget. It does not manufacture a confidence interval.

5. Record pressure and power. Measure each filter-path pressure drop separately using reviewed static tap positions. Do not drill before sealing/reach review. Record total electrical input with a suitable rated instrument; 16.8 W nominal fan load is not measured wall power.

Particle trial: useful, not certification

Prepare a controlled room. Record room dimensions, sensor positions, ventilation/HVAC, doors/windows and occupancy. Compare sensors together beforehand; document their response, limitations and detection floor. Keep locations and room conditions comparable across runs.

Do not generate smoke or dust yourself. No incense, combustion, powders or deliberate pollutant release in an occupied room. A formal challenge-aerosol test belongs with an equipped competent service. If ambient particles are too low or unstable, mark the trial inconclusive.

Suggested engineering trial: paired device-OFF baseline and device-ON observation windows, repeated under comparable conditions, using naturally present particles. Log raw PM, temperature and humidity; do not choose only the best-looking interval. Duration must provide adequate signal while conditions remain stable; no invented universal duration.

Analysis boundary: a PM decrease may reflect settling, ventilation, changed sources or sensor humidity response. A simple before/after percentage is not device efficiency. Blank particle data are supplied, but no automatic CADR calculator is issued before the room method and uncertainty are defined.

What can be reported: dated observations, run conditions, measured flow/power, particle trends and limitations. Formal CADR needs a validated method; HEPA-level efficiency needs appropriate filter/installed-leak evidence. Airflow alone proves neither.

No outdoor bubble claim: indoor results cannot establish a public-space radius or tower spacing. Heat, weather, wind, public access, stability and maintenance are separate design and test problems. Carbon/gas removal is outside this particulate trial.

Acceptance, tools and handoff

Before testing: rated protective design and qualified commissioning; guard/access/strength/seal closure; calibrated instruments and a reviewed test method. Funding-only support means rent instruments or commission an equipped service; no college lab is assumed.

Instrument functions, not purchase orders: total-flow measurement suitable for low-pressure fans; differential pressure measurement resolving expected small losses; rated input-power measurement; temperature/humidity logging; comparable particle monitors; noise meter if reporting noise. Exact ranges/accuracy and availability must match the approved method.

Acceptance rules: safety faults mean STOP. Missing method, invalid calibration, leaks, sensor saturation, unstable room conditions or insufficient particle signal mean INVALID/INCONCLUSIVE, not PASS. Flow below a project target requires design review; the 300 m3/h investigation point is not a safety or certification threshold.

Reproducible tools: run `scripts/analysis/d01_validation.py` for synthetic tests. With accepted physical data: `add --airflow-csv FILE --output NEW_FILE.json`. Existing output files are not overwritten. Blank templates, NaN, missing evidence, duplicate cells and mixed runs are rejected.

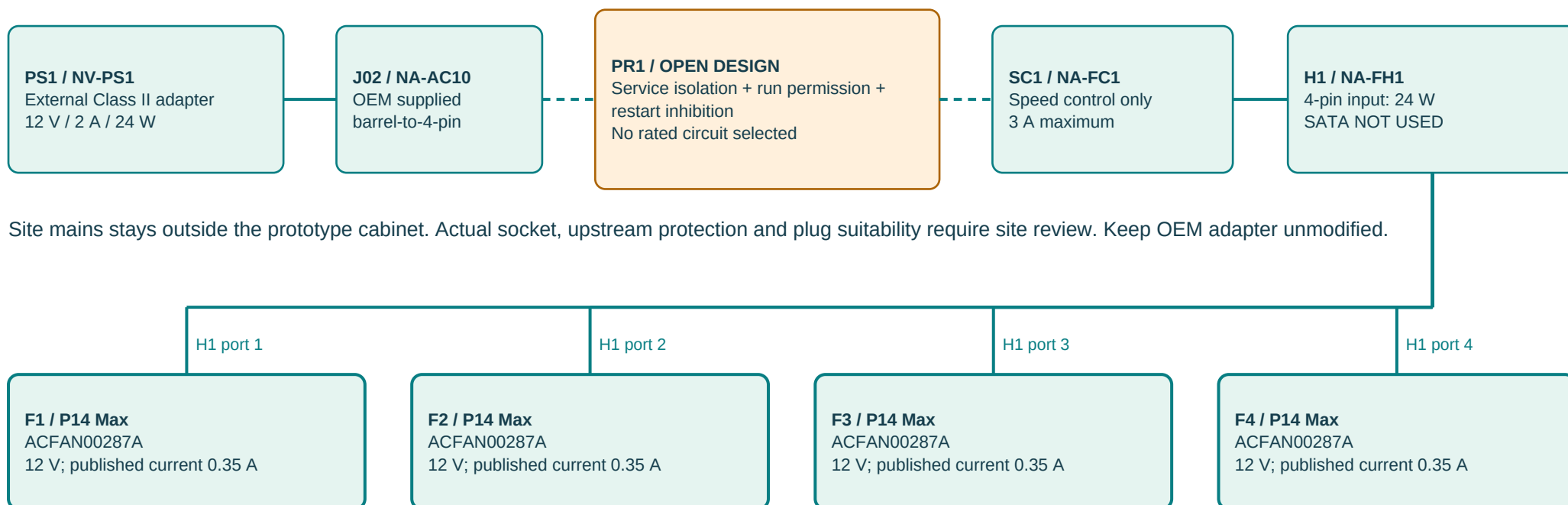
Checks completed: 15 synthetic/arithmetic checks. Input hashes link the pressure screen to the OEM CSV and current guard pattern. No physical measurements or tests performed; no purchases, installation or supplier messages.

Three packages now exist, not three approvals: mechanical R03M; electrical E00; validation V00. Remaining critical path is actual protective/structural/interface closure, qualified review, authorized build, commissioning and measured performance. More PDFs cannot substitute for those steps.

Sources: archived ARCTIC numeric workbook (source URL/hash in JSON); R03M `guard_pattern.json`; [EPA Guide to Air Cleaners](#) and [EPA DIY Air Cleaners](#), checked 4 October 2026. EPA guidance is not approval of D01. Test procedures here are proposed project methods, not copied certification standards.

One power path, four fan branches

OEM reference connectors are named; amber functions are unselected. No bridge across PR1 is permitted.



Site mains stays outside the prototype cabinet. Actual socket, upstream protection and plug suitability require site review. Keep OEM adapter unmodified.

OEM interfaces: SC1 output to H1 4-pin input using compatible OEM cable; H1 ports1–4 to individual fans. Lines represent connector-level power/control bundles, NOT conductor routing or plug orientation. PR1 insertion would require a designed compatible harness; no OEM cable should be cut to implement this sketch.

Only F1 RPM is forwarded upstream. Other port LEDs are local indications, not independent safety outputs. Do not tie tach outputs together. A lamp or zero PWM does not prove isolation or standstill.

Release hold: the pictured OEM chain alone does not implement the required STOP / deliberate RESET / fresh START behavior. PR1, cable entry, enclosure, conductor protection and all physical connections remain unreleased.

Load and thermal constraints

Catalogue limits are verified. Startup multipliers and auxiliary loads below are illustrative assumptions.

Path item	Published limit / value	Engineering interpretation
Four P14 Max fans	4 x 0.35 A = 1.40 A; 16.8 W at12 V	Published fan current; not measured startup or guaranteed maximum draw
NV-PS1 external adapter	12 V; 2 A /24 W	Governs current together with selected hub input
NA-FC1 speed controller	3 A /36 W maximum	Rating is NOT its own consumption; auxiliary load unknown
NA-FH1 selected input	24 W via4-pin; 2 A at12 V	54 W SATA rating does NOT apply to this path
Temperature	Fans0–40 C; adapter0–40 C; hub-40–60 C	Controller/enclosure environment unresolved. Not whole-device outdoor approval

Assumed fan-current multiplier	Assumed auxiliary load	Total demand	Difference to2 A
1.00x	0.00 A	1.40 A	+0.60 A
1.00x	0.10 A	1.50 A	+0.50 A
1.25x	0.10 A	1.85 A	+0.15 A
1.50x	0.00 A	2.10 A	-0.10 A
2.00x	0.00 A	2.80 A	-0.80 A

Consequence: with0.10 A assumed auxiliary load, the arithmetic fan-current multiplier ceiling is1.357x. Actual startup magnitude/duration is UNKNOWN. Exceeding a continuous catalogue limit does not predict the actual trip timing. Conversely, positive arithmetic margin is not a safety pass.

Do not simply install a stronger adapter: a5 A adapter feeding the same2 A hub path still leaves2.8 A hypothetical fan demand above the hub limit. A changed power architecture requires new connectors, protection and recovery analysis. Do not parallel adapters or connect SATA in this design.

Fan pin functions and wiring limits

OEM pin function reference only. Connector view, colour mapping and the physical terminal schedule are NOT released.

P14 Max pin	OEM function	Design restriction
1	Ground (-)	Signal/power return is not protective earth. Do not infer connector view or wire colour.
2	Power (+)	12 V reference. Rated switching and conductor protection unselected.
3	Tachometer output	Do not connect multiple tach outputs together or assume a safe-standstill signal.
4	PWM control	Speed command only; no claimed protective stop or service isolation.

Voltage-drop calculation: $\Delta V = I \times 2L \times \rho / A$. At assumed copper $\rho_{20} = 0.0175 \text{ ohm mm}^2/\text{m}$, one-way 1 m, 0.5 mm² and 1.4 A: $\Delta V = 0.098 \text{ V}$. At 40 C with assumed $\alpha 0.00393/\text{K}$: 0.106 V. These are examples, NOT the OEM cable gauge or a selected wire.

Coordination input	Current status	Needed to release
Fan terminal voltage	Allowed running range UNKNOWN	Do not use published 3.9 V startup voltage as allowed running minimum
Wire / connectors	Actual conductor gauge, contact rating, resistance UNKNOWN	OEM evidence plus routing, temperature, retention and measured terminal voltage
Fuse / protection	Hub auto-reset fuse; time-current / current-limit curves UNKNOWN	Coordinate fault energy with weakest conductor/contact and rated DC device; no arbitrary 2 A fuse
Switching / enclosure	PR1 and enclosure UNSELECTED	Actual DC load/inrush ratings, compatible harness, cooling and service/isolation arrangement

Keep the logger independent: no GPIO, USB or laptop feed into the fan-power or protective path. OEM PSU Class II does not decide the complete prototype protection class or whether any conductive parts require bonding; assess the actual assembly.

Protection coverage and release evidence

Digital work defines the gap; it does not fabricate a protective relay or claim physical commissioning.

Event / hazard	OEM chain evidence	Remaining implementation
Mains lost / restored	No manual-start latch demonstrated	Fresh start / held-start rejection not implemented
Hub overload recovers	Input fuse auto-resets	Recovery coverage must be resolved without defeating OEM protection
Fan2 /3 /4 fails	Local LEDs; only fan1 RPM forwarded	No independent four-fan protective monitoring claimed
Service / guard removal	No rated isolation assembly selected	Prevent reconnection; verify run-down and inaccessible impellers
PWM/monitoring failure	No stopped-state guarantee	Risk allocation, loss-of-signal test and separate physical protection

Scope to close, not a generic review: (1) determine actual reachable hazards and accept protection/restart requirements; (2) select rated PR1, all supply sensing and DC switching together; (3) establish wire/connector and protection coordination from measured/OEM current/time data; (4) integrate enclosure, protected entries and service isolation; (5) witness actual power restoration, held START, reset, fault recovery and safe-access cases. Existing restart rules are unchanged.

Deliverables included: POWER_COORDINATION.json has 20 load and 36 wire-drop scenarios; scripts/analysis/d01_electrical_closure.py reproduces them. Ten analytic/synthetic tests check arithmetic and that no synthetic result grants energization. Physical prototypes/tests remain 0. No component purchases, live fault tests or invented India stock/prices.

Primary references checked 5 October 2026: ARCTIC P14 Max specification sheet; Noctua NV-PS1 / NA-FC1 / NA-FH1 specifications and NV-PS1 overview; NA-FH1 features. Exact links are in POWER_COORDINATION.json. HSE maintenance guidance is used only for isolation principles, not Indian law or device compliance.

Recommendation: retain the supervised indoor OEM-chain reference. Do not energize this incomplete circuit or bypass PR1. A qualified reviewer must resolve the specific physical/electrical decisions above; the assistant supplies the calculations and drawing evidence, not a false safety approval.